



Quantum Mechanics

reference guide

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*The principles of reality are atoms, and emptiness, while other things are mere
opinion. (Democritus)*

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0.1 Preface

The *cover* is a figure of the infinity symbol. These notes takes broad studying to comprehend.

Abstract Mathematics(models of inductive reasoning), is availed to avoid nonsensical answers, and bogus non-differentiable functions. This is the task for the human mind; to see this truth, and preserve it. It's accepted that the Angular Momentum of a closed system is conserved(hence production of particles by conversion of gravitational energy into matter energy, conditioned by Biological factors). The origin of the universe(a Physical picture of a polarizable space, and characterized as young by $\frac{\epsilon}{\epsilon_0} - 1$, with an energy density when expressed in Newtonian units), is not a reproducible event, and cannot be duplicated in the laboratory. In trying to eliminate the Riemann curvature tensor, there's also the Ricci curvature tensor, assuming the photon travels without a shift in energy or frequency(red shift). There are formidable problems facing String Theory: it has no predictive power, it's a highly sophisticated theoretical laboratory(rather than experimental), it's defined at the Planck energy(10^{19} GeV), as any Physical theory that you can possibly come up with, it cannot be tested with the present technology, and it's tightly constrained as much as the graviton emerges as a massless state with spin-2. Space-time is beyond being verified experimentally, and of the vacua states, only one is needed to be isolated, and the Cosmological constant is very close to 0. However, in this theory, when there's spontaneous symmetry breaking, one does not know how to keep the Cosmological constant 0, perhaps a solution will be found to isolate the one vacuum.

No space/time for boredom! Have fun, and ask a Physics lab fundamental questions.

The letters in bold transform like vectors!, and the entire notebook parses on TeX Live 2023/Debian kpathsea version 6.3.5, jupyter notebook/lab, and ipynb viewer(Android).

These are notes/references, few functional code!, and compiling.

AN electron plays a central role, and can be assumed to move in a spherically symmetric field, where all binding energies are negative. These are advanced core notes(reference sheet) in Quantum Mechanics(describing how particles behave like waves - the state of a given system[internal structure], and its' probabilities), and some equations of Mathematical Physics[some theorems are mentioned to emphasize the concept of unitarity]. The key important formulas are the Schrödinger equation, Uncertainty Principle, energy levels of the hydrogen atom, harmonic oscillator energy, and the Hamiltonian energy(of which describes how a system evolves in time). Included, are physical quantities, long expressions, Gauge Theory, important systems(like the Harmonic Oscillator, the energy levels of the Hydrogen atom), Mathematical tools, important inequalities, and some advanced notes on Quantum Field Theory. The multiplication by i makes it possible to carry over to QM(Quantum Mechanics)...this is one of the most striking features of Classical Mechanics(CM)-namely, the correspondence between observables, and one-parameter groups of symmetries. We hereby have a Mathematical model capable in principle, of supplying the answer to any Physical question we might ask. The fact that the eigenvalues check with the observed spectrum of the hydrogen atom, when one takes $\mathbf{K} = \frac{\hbar}{2\pi}$, which provides a confirmation of the identity of $\hbar$, with $\frac{\hbar}{2\pi}$. Your objective, dear reader, is to show in theorems of analysis, providing asymptotic formulas for the eigenvalue distribution of differential operators. Don't try to read these notes linearly, instead, come back here as a formula reference.

0.2 List of Figures in Chapters

1. Binary Entropy Diagram
2. Vector pointing towards the Sun
3. Quantum Harmonic Oscillator
4. Anharmonic vs simple Harmonic Oscillator
5. Relativistic Mass increase
6. Ground state search
7. Sampled wave-function distribution

8. Breit-Wigner distribution
9. Stern-Gerlach projection

0.3 Diagnostic

(a) Show that $y = 2x^2$ is a solution to the differential equation: $\frac{1}{2}x^2y'' - xy' + y = 0$,

(b) If $y_1(t) = e^{3t}$, and $y_2(t) = e^{-3t}$ are solutions to $y'' - 9y = 0$, what is the general solution? Note that y_1 and y_2 are NOT constant multiples of one another, so they are linearly independent. Thus the general solution in the differential equation is
 $y(t) = c_1e^{3t} + c_2e^{-3t}$.

(c) If $y'' - 2y' + 5y = 0$, has the associated characteristic equation $\lambda^2 - 2\lambda + 5 = 0$, by the quadratic formula, the roots of the characteristic equation are $1 \pm 2i$. Therefore, the general solution to this differential equation is $y(x) = e^x(c_1\cos 2x + c_2\sin 2x)$.

(d) Knowing when to apply the **Product and Chain Rules, Total and Partial Derivatives**.

NOTE: $\frac{\hbar}{i} = -i\hbar$ (Multiplying by -i is the same as dividing by i.)

PROCEED



Note

Elementary logic functions(Boolean gates), or Disjunctive Normal Forms, Computation complexities, Algorithms, and Quantum circuits, are not part of this material.

• *Schrödinger Equation: General linear homogeneous solution(from translation operators), to the time-dependent Schrödinger Equation $\longrightarrow$*

$$\psi(x, t) = \sum_{n=1}^{\infty} C_n \psi(x) e^{-\frac{iEt}{\hbar}}, \quad t = 0 \quad (1)$$

• *Integral form, time-dependent $\longrightarrow$*

$$\psi(x, t) = \int \mathbf{K}[x, t : q_0, t_0] \psi(q_0, t_0) dx_0, \text{ where } \mathbf{K} \text{ is the kernel or propagator.} \quad (2)$$

Schrödinger Time Independent Equation

$$\boxed{E\psi_0(x) = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + V\psi_0(x)} \quad (3)$$

• *The identity Zero: Zero has no multiplicative inverse!!*

• *Informational Warning: All descriptions here are in the literature, and not original. DO NOT attempt to frame a physical picture!!, or take Relativity into sufficient account. Quantum Field Theory follows once these principles are mastered.*

• *On Rational numbers: If $(a) \neq 0$, there is no rational number($\mathbb{Q}$), where $(b) \times 0 = (a)$, can be fulfilled.*

Chapter 1 Familiar, Recognized, And Understood Equations

If $\frac{d\alpha}{dt} = 0 \forall t$, then α is constant

$$(f, \phi,) = \lim_{\alpha \rightarrow 0} (f, \phi_n) = 0$$

$$\frac{d\sigma}{d\Omega} = |f(\theta, \phi)|^2$$

$e^{i\mathbf{a}\cdot\mathbf{X}}e^{1\mathbf{b}\cdot\mathbf{P}}\phi_0$ is also a minimum uncertainty state

$$a^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle$$

$$\hat{j}_{\pm} \mid j, m\rangle = \hbar\sqrt{j(j+1)-m \mid m\pm 1)} \mid j, m\pm 1\rangle \quad (1.1)$$

$\implies$

Angular-momentum identity/Discrete orientation, with Quantum-number. (In units of $\frac{\hbar}{2\pi}$).

$$A(t) = e^{+\frac{iHt}{\hbar}} \mathbf{A} e^{-\frac{iHt}{\hbar}}$$

$$\hat{\mathbf{H}} = p\hat{q} - \mathbf{L}$$

$$ct' = \gamma(ct - \beta x)$$

$$E = (\mathbf{p}^2 c^2 + m^2 c^2)^{\frac{1}{2}}$$

$$\mathbf{E} = -\frac{\dot{\mathbf{A}}}{c} - \nabla\varphi$$

$\forall F \in HL^2(\mathbb{C}, \mu_{\hbar})$, we have:

$$\begin{aligned} \|T_{\mathbf{a}}F\|^2 L^2(\mathbb{C}^n, \mu_{\hbar}) &= (\pi\hbar)^{-n} \int \mathbb{C}_n e^{-\hbar|\mathbf{a}|^2} \mathbf{e}^{-2\mathbb{R}(\bar{\mathbf{a}}\cdot\mathbf{a})} |\mathbf{F}(\mathbf{z} + \hbar\mathbf{a})|^2 \mathbf{e}^{-\frac{|\mathbf{z}|^2}{\hbar}} dx = (\pi\hbar)^{-n} \int \mathbb{C}_n e^{-\frac{|\mathbf{z}+\hbar\mathbf{a}|^2}{\hbar}} |\mathbf{F}(\mathbf{z} + \hbar\mathbf{a})|^2 dz \\ &= \|\mathbf{F}\|^2 L^2(\mathbb{C}^{\times}, \mu_{\hbar}) \end{aligned} \quad (1.2)$$

$\prod(e^X) = e^{\pi(X)} = \sum_{x=0}^{\infty} \frac{\pi X^m}{m!}$ (Each element is a product of Lie algebra elements. Its power series is finite dimensional, and closed.)

The Hilbert-Schmidt-norm of a matrix $X \in M_n(\mathbb{C}) \longrightarrow$

$$\|X\|_{HS}^2 = \sum_{j,k=1}^n |X_{jk}|^2 \quad (1.3)$$

Cauchy-Schwarz inequality

$$\|XY\|_{HS} \leq \|X\|_{HS} \|Y\|_{HS} \forall X, Y \in M_n(\mathbb{C}) \quad (1.4)$$

$SO(3) \longrightarrow$

$$\sigma_\ell = \pi'_\ell \circ \phi^{-1} = \pi_\ell \quad (1.5)$$

$\hat{\otimes} \rightarrow$ Hilbert tensor product

Feynmann Path-Integral $\rightarrow$

$$e^{-\frac{i t \hat{H}}{\hbar}} \psi(X_0) = C \int_{\text{paths with } X(0)=X_0} \exp \left\{ \frac{i}{\hbar} (x(\cdot), 0, t) \right\} \psi(x(t)) Dx \quad (1.6)$$

Gauge-transformation $\rightarrow$

$$U_\gamma Q_{pre}^1(f) U_\gamma^{-1} = Q_{pre}^2(f), \quad \text{where } (U_\gamma) \text{ is the map, and "pre" is for pre-quantization.} \quad (1.7)$$

$$a' = a - \nabla \psi \quad \phi' = \phi + \frac{1}{c} \frac{\partial \psi}{\partial t}$$

This was David Bohms' view of the character of Electromagnetic potentials.

$$\psi_n(r, \phi) = f(r) e^{-in\phi}, \quad \text{where } n \text{ is an integer, and } f \text{ is an arbitrary holomorphic function on } (C), \text{ with } \int_0^\infty |f(r)|^2 r dr < \infty \quad (1.8)$$

$$\text{Symplectic-potential} \rightarrow \theta = \frac{1}{2}(p dx - x dp) = \frac{1}{2m\omega}(p dy - y dp)$$

$$\text{Fourier-transform}(\mathcal{F}) \rightarrow A \text{ constant multiple of a unitary map} \rightarrow \psi(p) = - \int_{\mathbb{R}} \phi(x) e^{-\frac{ixp}{\hbar}} dx dp$$

$$C_c^\infty(\mathbb{R}^\times) \rightarrow \text{Space of smooth, compactly supported functions}$$

1.1 Common nomenclature

1.1.0.0.1 Nomenclature

1. Written in polar coordinates, ρ and θ are in the unit circle; When $\rho < 1$, we have the series $u = \sum_{\nu=1}^{\infty} a_\nu \rho^\nu \cos \nu \theta$
2. $\pm$ means $-1^{|r|}$
3. ∇ - nabla Differential operator in ordinary 3D space
4. ∇h - Gradient of scalar h
5. $\nabla \cdot \mathbf{A}$ - Divergence of a vector
6. ∇^2 - Laplacian operator in ordinary 3D space spacetime $\rightarrow \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$
7. $\square$ - nabla operator in 3D Minskowski spacetime
8. $\square^2$ - d'Alembertian operator in Minskowski 4D spacetime
9. $\overline{V} = (V^\perp)^\perp$ is the closure of V
10. $g_{i,j}, g^{i,j}$ - Covariant, contravariant metric tensor of 3D space, or its components
11. ϵ_0 - Permittivity of free space
12. μ_0 - Permeability of free space
13. ρ, ϕ, z - Cylindrical coordinates in 3D space
14. σ - Spacetime interval in 4D
15. ϕ - Electromagnetic scalar potential
16. ω - Angular-frequency/speed

1.1.0.0.2 Structure

The content of the book is based on spaced structuring, and iterative deepening.

I would like to appreciate L^AT_EX , and



for making "learning coding" possible, for industry standards.

Chapter 2 Physical Ideas That Captivate

2.1 Characteristics of describing Energy

There is NO electronic ink for philosophical arguments! i.e. local realism does not exist, how long will $\mathbb{Z}$ remain a bachelor, cold fusion, climate change, singularities, why wave-functions overlap, determinism, hidden variables(proof that subquantum level don't exist), why periodicity of quantum numbers lead to periodicity of chemical elements, spin-flip transitions(oscillating magnetic monopoles/distance transverse magnetic $\mathbf{B}$ produced by an oscillating magnetic dipole, etc.... Ultra-cold atoms(lock-stepped with the same spin), have been realized in the laboratory(Bose-Einstein condensate[BEC]), through observing the center of mass system(a trick) of wave-like properties of atoms, so that the Pauli exclusion principle is NOT violated. This technology, although limited(interaction with the external environment depolarizes particles) has applications in quantum computers(cryptography - quantum key distribution ; Refer to figure 2.1), and is realized only for very low temperature Kelvin scales, not for classical computers. Even the speed of light (c) - has been stopped!, using slow light pulses. The wave-particle duality(using a super-computer) was proved by an experiment in 1974 by John Clauser(concludes that the wave-function collapse is an objective $\mathbb{R}$ process - not a physical process, and not Relativistic Invariant - agreed). The idea of measurement of the wave-function with any apparatus(alters the azimuthal quantum number/angular momentum of the orbital plane - resulting to a breakup of eigenvalues into a one-dimensional representation cylindrical symmetry of $2j + 1$, from spherical symmetry of j . Why, currently no answer/solutions...maybe its' even a wrong question. Considering Faradays' Law, knowing the Fourier transform, invoke a linear and orthogonal transformation to Stokes' theorem, generalizing how the electric field remains constant, the magnetic field too ($\text{curl}\mathbf{E} = -(\frac{\partial\mathbf{B}}{\partial t})$), with $\iint_s \mathbf{F}dS = \int_c \mathbf{F}dt$, where C is a closed curve, and $\mathbf{F}$ is a vector field defined in C . It can be calculated by making the electric field constant, and varying the magnetic field e.g. $\mathbf{B}(t) = \langle tx, ty, -2tz \rangle$, $0 < t < \infty$. Note that in general, the z component of angular momentum is $\hbar m$, with probability $\mathbb{K}$, and (m) is the state of the magnetic quantum number. This analysis is ficticiously true, and is based on accepting the theory of of complex numbers, Maxwells' equations, Fourier transform, homogeneous linear differential equations, the double-slit experiment(the results on the screen is a transmitted split beam intensity profile), Stern-Gerlach experiment(represents an entangled magnetic moment of Silver atoms), and Schrödingers' equation. There are no visible sources of some forms though, they're(strange and arbitrary) schematic representation of Physical ideas(Force, Momentum, Energy, Mass, Work, Acceleration. Length Of Interval, Time, and Velocity)...are all approximations! The simultaneous probability distributions of various observables, given differential equations, which may be integrated, show how these observables change in time. Keep in mind, we've evolved from thinking the Sun was static, (It is a linear combination of a stationary state[not static], its' probablities don't change!!, and that it revolved around the Eartha linear combination of a stationary state, with no changing probabilities), it's now the other way around.

. For any questions, suggestion or comment, feel free to contact me on GitHub [issues](#) or email me at stowino21@gmail.com.

2.1.0.0.1 Axioms of Quantum Mechanics

In Dirac notation, an element of space state is a Ket($|\ \rangle$). Time at t_0 is defined by specifying a Ket $|\ \Psi(t_0) \rangle$, belonging to a Vector space.

1. **A1:** The wave-function is the complete description of a quantum system of an abstract vector space(state space ϵ).
2. **A2:** Hermitian operators are observables.
3. **A3:** The measurement axiom; A measurable physical quantity A , is described by an observable A , acting on ϵ .
4. **A4:** Time evolution via the Schrödinger equation.

2.1.0.0.2 Illustrating Vectors

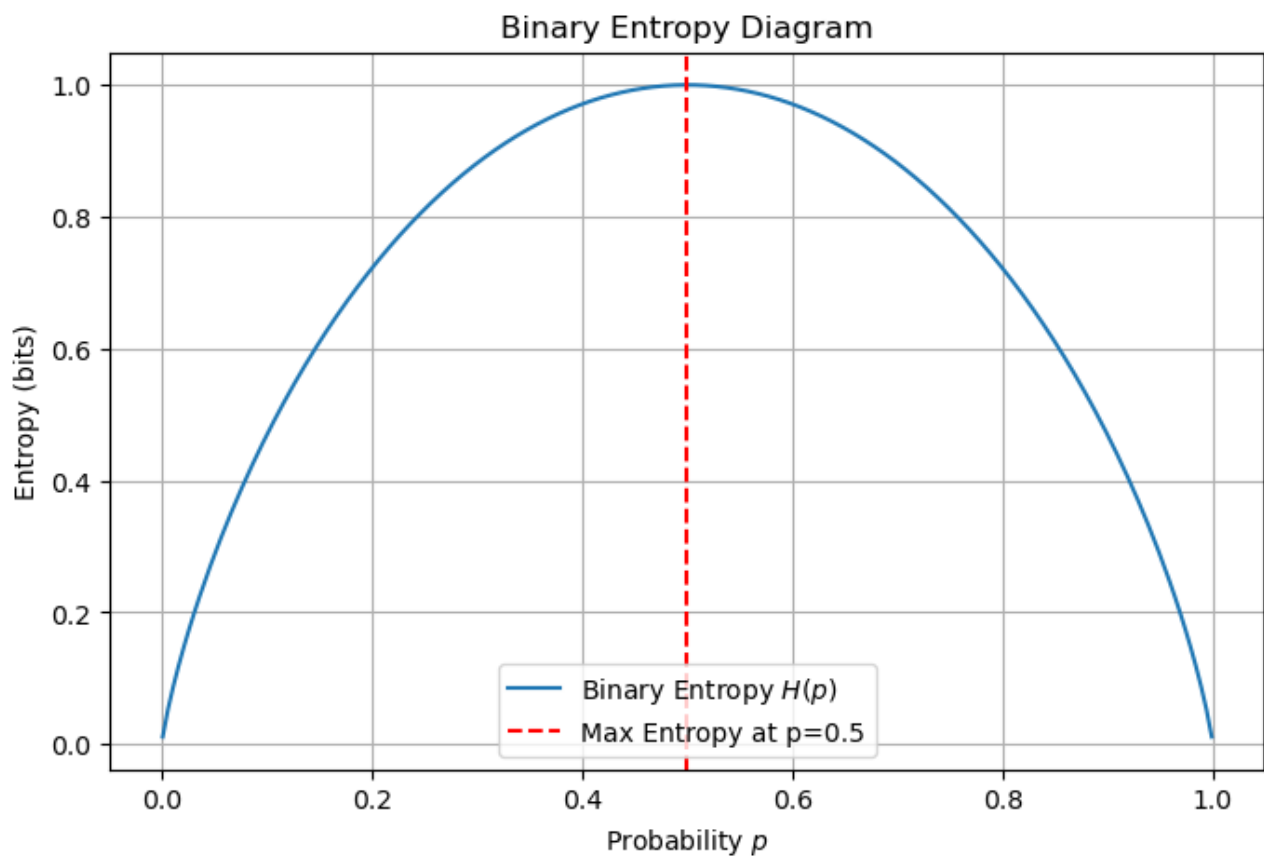


Figure 2.1: Binary Entropy

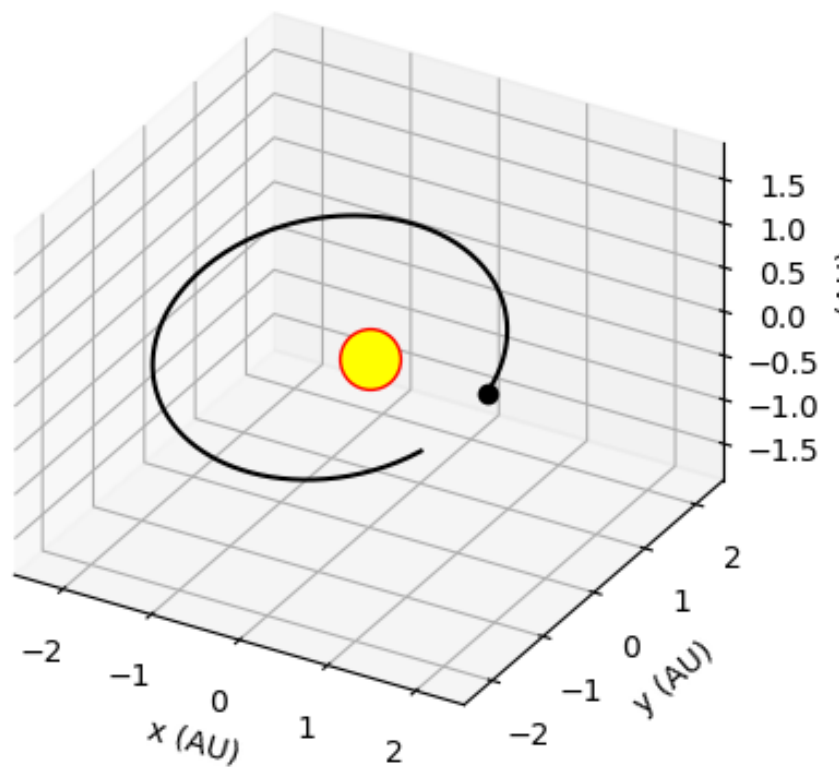


Figure 2.2: Vector pointing towards the Sun

2.2 Some Conceptual Frameworks; The names in color are Nobel prize receipients, and nomenees.

- Copenhagen interpretation(s) - Bohr, Heisenberg, Pauli, von Weisäcker, Peierls.
- Non-local Hidden Variables - de Broglie, Bohm, Maudlin, Goldstein, Hardy.
- Convival Solipism - Zwirn
- Modal Interpretations - Kochen, Dieks, van Fraassen, Bub, Sipe.
- Conscienceness-induced Collapse - Wigner, von Neumann, Josephson.
- Objective Collapse Models - Ghirardi-Rimini-Weber, Pearl, Penrose, Tumulka.
- Consistent/Decoherent Histories - Griffiths, Omnés, Gell-Mann, Hartle.
- Superdeterminism - t'Hooft, Hossenfelder.
- Rational Interpretations - Rovelli
- Ensemble Interpretations - Ballentine, Einstein

- Informational Interpretation - [Zellinger](#), Brukner, Bulb.
- Many-Worlds Interpretation - Everett, Hawking, Science-fiction fans.

Remark

Required reading:

- Spectrum of the harmonic oscillator
- Spectral Theorem
- Linear Algebra(particularly Matrix Theory)
- Continuous Functional Calculus(systems of 2nd order differential equations for vector-valued functions, and factoring)
- Measure Theory
- Equi-partition theorem(law of equi-partition of Energy)
- Electromagnetic theory of light.

Chapter 3 Terminology

3.1 Definitions

Quantum-Theory $\rightarrow$ A generalization/systematic refinement of classical mechanics

Quantum-communication/computation $\rightarrow$ Manipulating qubits with quantum gates; Qubits being transmitted between different parties; Exploring applications based on sending information, rather than processing it.

Quantum Fourier Transform $\rightarrow$

$$\phi = \sum_{k=0}^{n-1} y_k |k\rangle = \frac{1}{\sqrt{2}} \sum_{k=0}^{n-1} e^{\frac{2\pi i j k}{N}} |k\rangle = \frac{1}{\sqrt{2}} \sum_{k=0}^{N-1} \omega^{jk} |k\rangle \quad \text{For } N=2, \quad k=0,1 \quad \text{and} \quad \omega = e^{\frac{2\pi i}{N}} \quad (3.1)$$

An-entangled-pair $\rightarrow$ Applying one qubit by manipulating the larger set of the four possible states.

Von Neumann entropy $\rightarrow$

$$S(\rho) = -\text{tr}(\rho \ln \rho) = -\sum_j \lambda_j \ln \lambda_j \text{ (These are diagonalized values); } S(\rho) = 0 \iff \rho^2 = \rho \text{ (case of a pure state/projector)} \quad (3.2)$$

Maximally mixed state $\rightarrow S(\rho_m) = -N \frac{1}{N} \ln \frac{1}{N} = \ln N$

Gibbs entropy(classical statistical mechanics) $\rightarrow$

$$S_G = -k_B \sum_i P_i \ln P_i, \text{ where } k_B \text{ is the Boltzmann constant, and } P \text{ is probabilities} \quad (3.3)$$

Shannon entropy(classical information theory) $\rightarrow$

$$H(X) = \mathbf{E}(\log_b P(X)) = -\sum_x P(x) \log_b P(x), \text{ where } H \text{ is for Shannon, } \mathbf{E} \text{ is the expectation/average value, } X \text{ is a variable, and } P \text{ is p} \quad (3.4)$$

State-reduction $\rightarrow$

The effective two-level Hamiltonian($\hat{H}_{eff} = \begin{pmatrix} 0 & \Omega \\ \Omega & \Delta - i\frac{\Gamma}{2} \end{pmatrix}$), dissipating via spontaneous emission, during "measurement", included in the dynamics(A Quantum trajectory is NOT reversible). Note that $[\hat{H}_{eff}, \hat{H}_{eff}^\dagger] \neq 0$, this Hamiltonian also generates a non-unitary dynamics $e^{-i\hat{H}_{eff}(t)} \neq e^{i\hat{H}_{eff}^\dagger(t)}$

Measurement-of-a-state(indicates a loss of energy) $\rightarrow$

$$\langle \psi | \psi \rangle = \langle \psi_0 | e^{-i\hat{H}_{eff}(t)} e^{i\hat{H}_{eff}(t)} | \psi_0 \rangle \rightarrow e^{-\Gamma(t)} \quad (3.5)$$

$\log_b$ can take any of the values 2[Classical bits], e[the Natural Logarithm], and Base 10[orders of magnitude]

$$H_{bin}(p) = -p \log p - (1-p) \log(1-p) \rightarrow \text{e.g. } H_{bin}(\frac{1}{2}) = -\frac{1}{2} \log \frac{1}{2} - \frac{1}{2} \log \frac{1}{2} = \log 2$$

Normalization - technical manipulation of Mathematical symbols(to eliminate the physical vacuum contribution, and to simplify the one electron contribution).

Vacuum-region $\rightarrow$ discontinuity of fields in the small.

See figure 3.2 Limit - $\forall$ total infinities, the degrees of freedom ceases to increase without limit as shorter and shorter time intervals are considered.

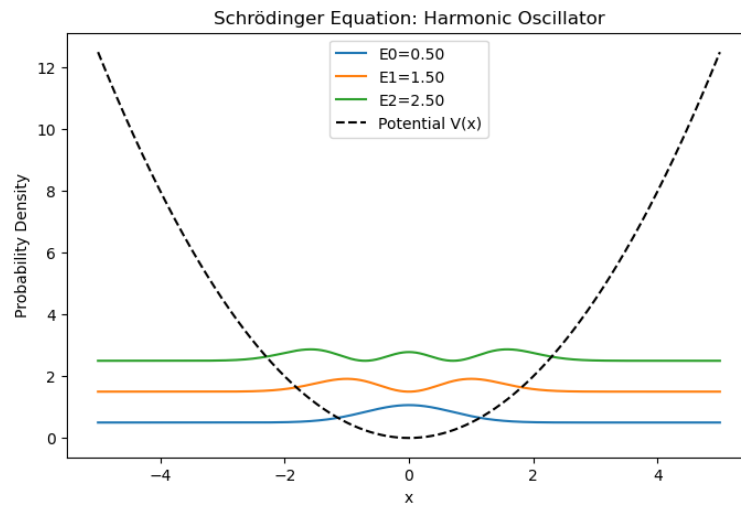


Figure 3.1: This is a figure of the Quantum Harmonic Oscillator. It is defined by $E_n = (n + \frac{1}{2})\hbar\omega$

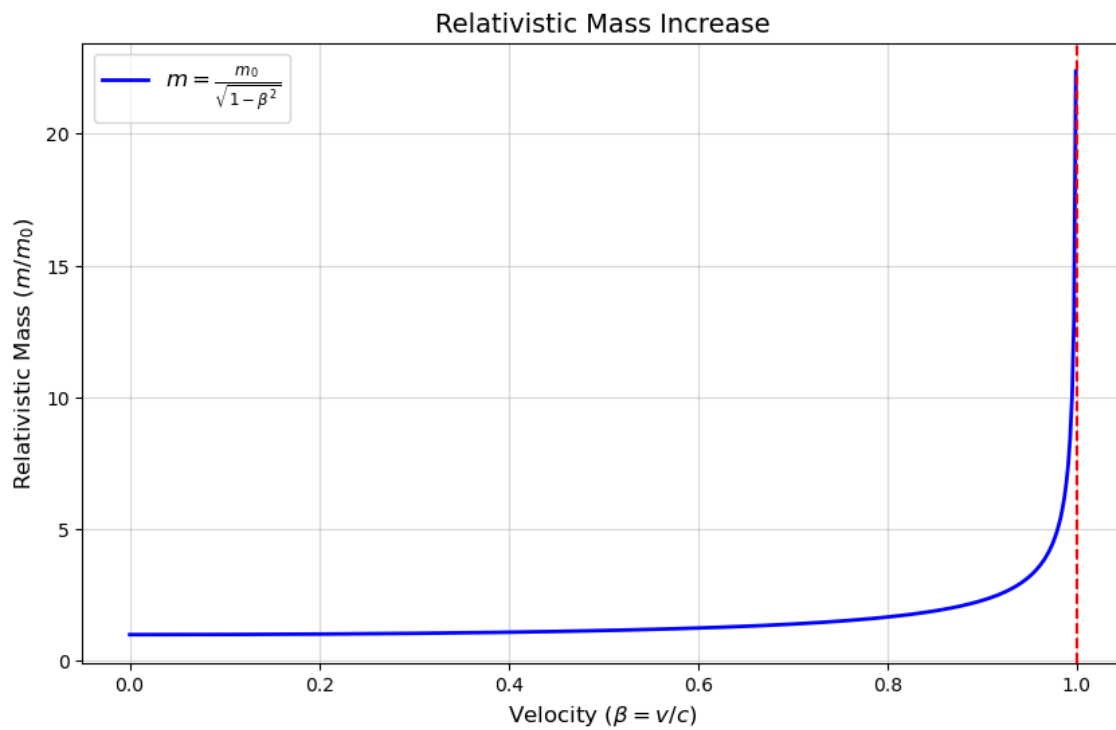


Figure 3.2: Limit concept of Mass with the Speed of Light

Wave-equation $\longrightarrow$

$$\frac{\partial^2 u(x, t)}{\partial t^2} = c^2 \frac{\partial^2 u(x, t)}{\partial x^2} [\text{Relates acceleration and curvature}]. \quad (3.6)$$

Decoherence - The inevitable interaction of a system and its environment(quantum-to-classical transition; Its a fundamental limiting factor in the useful operation of Quantum Computers).

Spin(total-angular-momentum)/directional quantization) - a particle ℓ of non-negative integer or half integer. The Hilbert space for such a particle is $\mathcal{L}^2(\mathbb{R}^3) \hat{\otimes} V_\ell$, where V_ℓ is an irreducible projective representation of $SO(3)$ of dimension $2\ell + 1$

Photons - A collection of an indefinite number of indistinguishable particles, that obey Bose-Einstein statistics, with integer spin $\mathbb{N}$.

Boson(Bose-Einstein statistics) - A particle with an integer spin

Fermion(Fermi-Dirac statistics) - A particle with half-integer spin e.g. an electron

For $j, k, l \in 1, 2, 3$ we define $\epsilon_{j, k, l}$ by the formula:

$$\epsilon_{j, k, l} = \begin{cases} 1 & \text{if } j, k, l \text{ is an even permutation of } (1, 2, 3) \\ -1 & \text{if } j, k, l \text{ is an odd permutation of } (1, 2, 3) \\ 0 & \text{if any two of } j, k, l \text{ are equal} \end{cases} \quad (3.7)$$

e.g $\epsilon_{3, 2, 1} = -1$ and $\epsilon_{2, 1, 2} = 0$.

The commutation relations for the basis F_1, F_2, F_3 for $SO(3)$ may be written(using the summation convention!) as:

$$[F_j, F_k] = \epsilon_{j, k, l} F_l$$

i.e. if we take $j=1$ and $k=2$, then the sum on ℓ gives a nonzero value only when $\ell = 3$, and we recover the relation $[F_1, F_2] = F_3$

Invariant $\rightarrow$ The density ν is irreducible under translations in the real directions.

Polarization - The set of directions in which the elements of the quantum subspace are covariantly constant.

Covariant-derivative $\rightarrow$ A connection ∇ operator for a vector field.

Integrability $\rightarrow$ Two complex vector fields X and Y lie in P_z at each point z , then so does $[X, Y]$

Hermitian-line-bundle $\rightarrow (L, \nabla)$ connection of $L^{\otimes k}$ having a curvature $\frac{\omega}{h}$ for any positive integer n (things behave nicely when k tends to ∞)

Half-form-space $\rightarrow$

$$s = (\psi \circ q) \otimes \sqrt{q^*(\beta)} \quad (3.8)$$

A measurable-function $\psi: X \rightarrow \mathbb{C}$ is said to be integrable, if $\int_X |\psi| d\mu < \infty$

Density of μ w.r.t. $\nu \rightarrow$

$$\mu(E) = \int_E \rho d\mu \quad \forall E \in \omega \quad (3.9)$$

supposing μ and ν are two σ -finite measures on a measure space (X, Ω) , and that μ is absolutely continuous w.r.t. ν : then there exists a non-negative, measurable function ρ on X .

Monotone-class $\longrightarrow$ Suppose $\mathcal{M}$ is a monotone class of subsets of a set X , and suppose $\mathcal{M}$ contains an algebra $\mathcal{A}$ of subsets X , then $\mathcal{M}$ contain the σ -algebra generated by $\mathcal{A}$

Dirac δ function $\longrightarrow$

$$\delta_x = \chi(0), \text{roughly} \quad \delta(x - X) = \begin{cases} \infty & \text{if } x = X \\ 0 & \text{otherwise} \end{cases} \quad (3.10)$$

The function illustrates the limit of a distribution.

If $\|\cdot\|: V \longrightarrow \mathbb{R}$, then

$$\|\psi\| = \sqrt{\langle \psi, \psi \rangle} \quad (3.11)$$

$\|\cdot\|$ is a norm on V

Hilbert-space $\longrightarrow$ A vector space $\mathcal{H}$ over $\mathbb{R}$ or $\mathbb{C}$, equipped with an inner product $\langle \cdot, \cdot \rangle$, such that $\mathcal{H}$ is complete in the norm above. [For a separable infinite dimensional $\mathcal{H}$, the partially ordered set of all questions in QM(Quantum Mechanics), is isomorphic to the partially ordered set of all closed sub-spaces].

Orthogonal-space $\longrightarrow$ If V is any closed subspace of $\mathcal{H}$, a subspace $V^\perp$ of $\mathcal{H}$, is:

$$V^\perp = \{ \phi \in \mathcal{H} \mid \langle \phi, \psi \rangle = 0 \quad \forall \psi \in V \} \quad (3.12)$$

Skew-self-adjoint $\longrightarrow$

$$A^* = -A \quad \text{One-parameter group of orthogonal/unitary transformation.} \quad (3.13)$$

Orthogonal-projection $\longrightarrow$ For a closed subspace V , where $V = \text{range}(P)$, P is the projection if it is any bounded operator on $\mathcal{H}$ satisfying $P^2 = P$, and $P^* = P$

A definite-state(random sequence) $\longrightarrow$

$$\|\bar{\sigma}_z^N \mid \psi\|^2 = \frac{1}{N^2} \langle \psi \mid \sum_{r=1}^N \sum_{s=1}^N \sigma_z^{(r)} \sigma_z^{(s)} \mid \psi \rangle \quad (3.14)$$

$$\langle \psi \mid \sigma_z^{(r)} \sigma_z^{(s)} \mid \psi \rangle = \delta^{rs} \implies \lim_{N \rightarrow \infty} \|\bar{\sigma}_z^N\|^2 = \lim_{N \rightarrow \infty} \frac{1}{N^2} N = 0 \quad (3.15)$$

Likewise for $\bar{\sigma}_z^a$ etc.

Zorns'-Lemma $\longrightarrow$

Let P be a set of elements $a, b, \dots$. Suppose there is a binary relation defined between certain pairs (a, b) of elements P , expressed by $a \prec b$, with the properties: $\begin{cases} a \prec a \\ \text{if } a \prec b \text{ and } b \prec a, \text{ then } a = b, \\ \text{if } a \prec b \text{ and } b \prec c, \text{ then } a \prec c \text{ (transitivity)} \end{cases}$ Then P is said to be partially ordered or semi-ordered.

Curl $\longrightarrow$ Measure of the rotation in the vector field about the points in the direction of the normal vector $\mathbf{N}$, and Stokes theorem justifies this interpretation.

Ampere's Law(line integral along a closed path) $\rightarrow \oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{encl}$, where μ_0 is the magnetic constant, and I_{encl} is the net current enclosed by the path.

Equi-partition-principle $\rightarrow$

$$E = \frac{3}{2}kT, \text{ where } k \text{ is the Boltzmann's constant.} \quad (3.16)$$

See page 45 for Ground-state $\rightarrow$ An electron in a stable innermost orbit/lowest energy state(most negative), and cannot fall into the nucleus.

Equilibrium position $\rightarrow$ An electron in a stable innermost orbit/lowest energy state(most negative), and cannot fall into the nucleus. (Its positive definite at its local minimum, hence its Energy is $\frac{\hbar\omega}{2\pi}$ at 0). The equality $\frac{\hbar\omega}{2\pi}$ is the Minimum Energy. $\psi_0(x) = (2\pi\sigma^2)^{-\frac{1}{4}} e^{-\frac{x^2}{4\sigma^2}}$, with $\sigma^2 = \frac{\hbar}{2m\omega}$, and $E = \frac{\hbar\omega}{2}$.

Pauli-exclusion-principle $\rightarrow$ No two or more electrons in an atom(Fermions), can have the same value, for any of the four quantum numbers.

Self-adjoint $\rightarrow T(F(\phi), \psi) = T(F(\psi), \phi) = T(\phi, F(\psi))$

Eigenvalue-equation $\rightarrow$ An equation, where the operator, operating on a function, produces a constant, times the function. [The function is called an Eigenfunction, and the resulting numeric is called the Eigenvalue]. Eigen here is the German word meaning self, or own.

Hamiltonian-energy(discrete) $\rightarrow$

$$\hat{\mathbf{H}} = \frac{1}{2}m\omega_c^2 \hat{r}_c^2 \quad (3.17)$$

This is the Kinetic Energy of a particle of Mass m with Cyclotron Frequency ω_c , and radius $\hat{r}_c$.

Dirac equation $\rightarrow$

$$i\hbar \frac{\partial \psi}{\partial t} = \left[c\boldsymbol{\alpha} \cdot \hat{\mathbf{p}} + \frac{e}{c}\mathbf{A} + \beta mc^2 + V(r) \right] \psi \quad (3.18)$$

Where the coupling of the electron to the scalar potential $\Phi(r)$ is included via $V(r) = -e\Phi(r)$.

Gibbs Inequality $\rightarrow H(f_1, \dots, f_n) \leq -\sum_{j=1}^m f_j \log(u_j)$

Projection-valued-measure associated with $f(A) \rightarrow f(A')$

[$f(A)$ is the self-adjoint operator, corresponding to the observable $f(A')$].

$E \rightarrow P_{f^{-1}(E)}^A$. Note that the operators corresponding to the Energy and Momentum observables of a quantum system are obtainable as infinitesimal generators of certain one-parameter groups of unitary operators.

Eigenvector of a self-adjoint operator $\rightarrow$ A state in which the corresponding observable takes on the corresponding Eigenvalues with probability 1

Von Neumann-density-matrix $\rightarrow$ The matrix of an operator defining a mixed state.

Differentiable-function ψ [Schrödinger's equation] $\rightarrow$

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} + \frac{\partial^2 \psi}{\partial z^2} + \frac{2m}{\hbar^2} \left(\mathbf{E} + \frac{e^2}{\sqrt{x^2 + y^2 + z^2}} \right) = 0, \text{ with } \hbar = \mathbf{K}. \quad (3.19)$$

Canonical-commutation-relation $\longrightarrow [\hat{X}, \hat{P}] = i\hbar\delta_{ij}, i,j = x,y,z.$

Heisenberg-commutation-relation $\longrightarrow$

$(X_{\phi_1}, X_{\phi_2} - X_{\phi_2}, X_{\phi_1}) = (\frac{\hbar}{i}) [f_{\phi_1}, f_{\phi_2}]$, where f_{ϕ} is an observable, and X_{ϕ} the self-adjoint operator.

Discrete $\longrightarrow$ A representation which can be decomposed as a direct sum of irreducibles.

Schurs' Lemma(only true in $\mathbb{C}$ representations, NOT $\mathbb{R}$ vector spaces) $\longrightarrow$ Let T be $\in R(L,M)$, then L restricted to the orthogonal complement of the null space of T , is $\equiv$ to M restricted to the Closure the Range of T .

Degeneracy $\longrightarrow$ Occurrence of multiple eigenvalues/states sharing the same energy.

Zeeman-effect $\longrightarrow$ Splitting apart of eigenvalues, hence of spectral lines, in a magnetic field.

Kronocker-delta(Orthonormality condition) $\longrightarrow \delta_{ij} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$

Tensor/Kronecker-product of two Spin one-half state space[needed to calculate the hyperfine-splitting in the Hydrogen-atom] $\longrightarrow$

$$|0,0\rangle = \frac{1}{\sqrt{2}} (|\uparrow\rangle_1 \otimes |\downarrow\rangle_2 - |\downarrow\rangle_1 \otimes |\uparrow\rangle_2) = \frac{1}{\sqrt{2}} (|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)$$

Clebsch-Gordan-coefficient(expansion)/vector coupling/Wigner $\longrightarrow$

$$|J, M\rangle = \sum_{m_1=-j_1}^{j_1} \sum_{m_2=-j_2}^{j_2} \langle JM | m_1, m_2 \rangle \langle m_1, m_2 | J', M' \rangle = \delta_{JJ'} \delta_{MM'} \quad (3.20)$$

Hydrogen hyperfine split $\longrightarrow$ Energy shift/difference of the lowest antiparallel state of the spin of the protons magnetic moment and the electrons magnetic field, pointing in the same direction(corresponds to the fine-structure splitting).

Collision-broadening $\longrightarrow$ Distortion of an emitted wave during radiation, thus smearing the spectral line.

See Eq. (4.5) **Inner-product** $\longrightarrow \langle A|B \rangle = \frac{1}{2} \text{tr}(A, B^\dagger)$

A Matrix changes a derivative, so $A = \frac{d}{dt}$. To find the transpose of this (A) we need to define the Inner Product between two functions $x(t)$ and $y(t)$.

The Inner Product changes from the sum of $X_k Y_k$ to the integral of $x(t) y(t)$.

Inner product of functions $X^T y = (X, y) = \int_{-\infty}^{\infty} x(t)y(t)dt$

From this Inner Product, we know the requirement on A^T . The word "adjoint" is more correct than "transpose" when we are working with derivatives.

The transpose of a matrix has $(Ax)^T y = x^T (A^T y)$. The adjoint of $A = \frac{d}{dt}$ has:

$$(Ax, y) = \int_{-\infty}^{\infty} \frac{d}{dt} y(t) dt = (x, A^T y)$$

Recognize integration by parts. The derivative moves from the first function $x(t)$, to the second function $y(t)$. During that move, a minus sign appears. This tells us that the transpose of the derivative is minus the derivative.

The derivative is antisymmetric: $\mathbf{A} = \frac{d}{dt}$, and $\mathbf{A}^T = -\frac{d}{dt}$. Symmetric matrices have $S^T = S$, antisymmetric matrices have $A^T = -A$. Linear algebra includes derivatives and integrals, because those are both linear.

This anti-symmetry of the derivative applies to centered difference matrices. $\mathbf{A} =$

$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & -1 & 0 & 1 \\ 0 & 0 & -1 & 0 \end{bmatrix}$$

;transposes to $A^T =$

$$\begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$= -\mathbf{A}$.

And a forward difference matrix transposes to a backward difference matrix, multiplied by -1. In differential equations, the second derivative (Acceleration) is symmetric. The first derivative (Velocity) is antisymmetric.

Entanglement $\longrightarrow$ A Bell state that cannot be expressed as a tensor product of a single qubit state with another single qubit state.

No-cloning-theorem $\longrightarrow$ One cannot create a copy of quantum states, although teleporting is permitted.

Teleportation $\longrightarrow$ Same quantum state in different locations.

See 3.1 for the Quantum-Harmonic-Oscillator $\longrightarrow$

$$\hat{H} = \hbar\omega \left(\hat{a}^\dagger \hat{a} + \frac{1}{2} \right) = \hbar\omega \left(\hat{N} + \frac{1}{2} \right), \text{ where } \hat{a} = \sqrt{\frac{m\omega}{2\hbar}} \left(\hat{x} + \frac{i}{m\omega} \hat{p} \right);$$

$$\hat{a}^\dagger = \sqrt{\frac{m\omega}{2\hbar}} \left(\hat{x} - \frac{i}{m\omega} \hat{p} \right), \text{ and}$$

$$\hat{N} = \hat{a}^\dagger \hat{a} = \frac{m\omega}{2\hbar} \left(\hat{x} - \frac{i}{m\omega} \hat{p} \right) \left(\hat{x} + \frac{i}{m\omega} \hat{p} \right)$$

Flux-quantum(a dimensionless quantity) $\longrightarrow \Phi = \frac{\hbar}{2e}$, where $2e$ is the charge of the electron(Cooper pairs).

Von Neumann-equation $\longrightarrow \partial_t \rho(t) = -\frac{i}{\hbar} [H, \rho(t)]$, where ρ is the density matrix, and H is the Hamiltonian; similar to the Schrödinger equation, $i\hbar \partial_t \rho(t) = [H, \rho(t)]$ (describes the time-evolution of the density operator).

$$\text{Resolution-of-an-identity} \longrightarrow \sum_{k_0} \sum_{\mathbf{k}_1} \langle k_0 | A_0 | \mathbf{k}_1 \rangle \langle \mathbf{k}_1 | A_1 | k_0 \rangle$$

3.2 Ideas

Ideas that work[linear, and angular momentum of a field

- A perfect crystal lattice(i.e. a One-Dimensional system) is effectively empty, for an electron in the lowest band, even though the space is full of atoms.

- The absolute value of Energy is equivalent to the fraction of a reduced mass of a proton-electron system, times Charge to the fourth power, over twice an integer squared times the square of the Plancks' constant $\longrightarrow |E| = \frac{\mu Q^4}{2n^2 \hbar^2}$

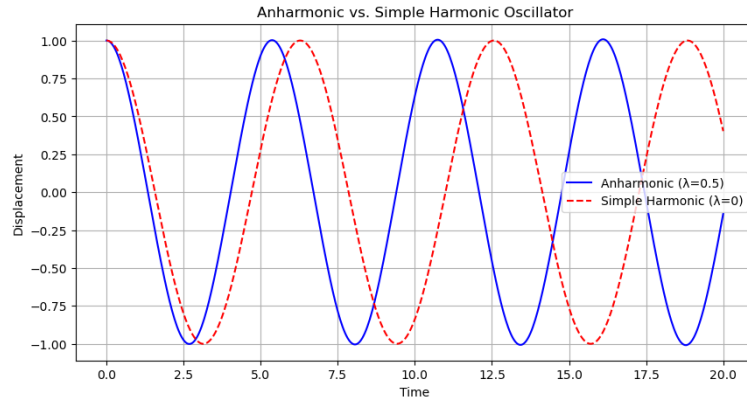


Figure 3.3: Representation of Anharmonic vs simple Harmonic Oscillator

- There are two special classes of polarizations(See 9), those that are purely real(i.e., $\bar{P}_z = P_z \forall z \in \mathbb{N}$), and those that are purely complex(i.e., $P_z \cap \bar{P} = 0 \forall z \in \mathbb{N}$).

- The half-form of the Hilbert space is the completion w.r.t. $|s|^2 = \int_{\Xi} \widetilde{(s1, s2)}$ of the space of polarized for which $|s|^2 < \infty$

- Pairing maps $\longrightarrow \int_{\mathbb{R}} \int_{\mathbb{R}} \phi(x) e^{-\frac{ixp}{\hbar}} dx \psi(p) dp$

- Dirac δ function $\longrightarrow \delta_x i = \xi(0)$

- Faradays' Law(The Electric field is the negative of the rate of change of the corresponding Magnetic field w.r.t time)
 $\longrightarrow \text{curl} \mathbf{E} = - \frac{\partial \mathbf{B}}{\partial t}$

Linear-system $\longrightarrow$ Its properties satisfy a finite-dimensional vector space, has a positive-definite quadratic form, and is independent.

Configuration-vector $\phi \longrightarrow \frac{d}{dt}(\psi, \phi) = \psi, -I^*(\phi)$ or $\frac{d\phi}{dt} = \psi \frac{d\psi}{dt} = -I^*(\phi)$, satisfies the second-order equation $\frac{d^2\phi}{dt^2} = -I^*(\phi)$, where ϕ is a scalar or vector-valued function on 3-space, and I^* is a differential operator.

- **Physical-law** (Seiberg-Witten-equations(useful on symplectic 4-manifolds)), is an example of a class of ϕs ''(a scaling-factor). proves one suitable existence and uniqueness theorems of the class of ϕs '' $\longrightarrow$ When ϕ is regarded as a function of x,y,z and t, then it becomes a partial differential equation of the form $\frac{\partial^2 \phi}{\partial t^2} = -I^*(\phi)$.

- Light $\longrightarrow$ Vibrations in the electromagnetic-field

Energy $\longrightarrow$ An observable corresponding to $\hbar$, times the dynamical operator, and that its an integral so that we have an **energy conservation law**.

- Heisenberg-commutation-relation $\longrightarrow Q_j Q_k - P_j P_k = i \hbar \delta_j^k$

- Wavefunction of a system $\longrightarrow$ A state vector ψ , is a square summable function on $\mathcal{M}$, and trajectory $e^{-iHt}(\psi)$ is a function on $\mathcal{M} \times \mathbf{R}$, where $\mathbf{R}$ is the real line(This is a function of $3n + 1$ real variable).

- Expected-value of the j-th Momentum component $\longrightarrow \frac{\hbar}{i} \int \left(\frac{\partial \psi}{\partial q_j} \overline{\psi} \right) dq_1 \dots dq_{3n}.$

Chapter 4 Physics of the Universe

4.1 Historical Timeline of the Quantization of Light

Table 4.1: Historical Timeline of the Quantization

Year	Description
1670–1678	Isaac Newton proposes the corpuscular theory of light, suggesting light consists of particles.
1801	Thomas Young performs the double-slit experiment, demonstrating the wave nature of light.
1864	James Clerk Maxwell formulates the theory of electromagnetism, describing light as an electromagnetic wave.
1900	Max Planck introduces the concept of quantization to explain blackbody radiation, proposing that energy is emitted or absorbed in discrete packets called <i>quanta</i> .
1905	Albert Einstein explains the photoelectric effect using Planck's quantum hypothesis, suggesting that light consists of discrete packets of energy called <i>photons</i> .

where:

$$E = h \nu$$

4.1.1 Key Takeaways

The quantization of light marked a paradigm shift in Physics, leading to the development of Quantum Mechanics. Light exhibits both wave-like and particle-like properties, a concept known as **wave-particle duality**.

4.2 Manifolds of pairing maps on an invariant self-adjoint subspace

4.3 Constants

A few constants:

$$\hbar \cdot c \simeq 197.327 \text{ MeV} \cdot \text{fm}$$

Coupling/Fine structure constant $\alpha \rightarrow \frac{e^2}{4\pi\epsilon_0\hbar c} \simeq \frac{1}{137.036}$, $m_e c^2 \simeq 0.511 \text{ MeV}$, $m_p c^2 \simeq 9.38 \text{ MeV}$. Described as deviations in the electron spin/removal of energy degeneracy of the orbital quantum number ℓ : $E = E_{n,\ell}$ [Energy shifts, and splitting of spectral lines].

$$\text{Bohr Radius} \longrightarrow r_B = \frac{4\pi\epsilon_0\hbar^2}{e^2 m_e} \approx 0.53 \times 10^{-10} \text{ m}$$

4.3.1 Useful Formulae

- (1) Relativity: $p = mv\gamma$, $E = \gamma m c^2$, $E^2 = p^2 c^2 + m^2 c^4$, $\gamma = \frac{1}{\sqrt{1-\beta^2}}$, $\beta = \frac{v}{c}$

- (2) de Broglie: $\lambda = \frac{h}{p}$,

Compton: $\lambda_C = \frac{2\pi\hbar}{mc}(1 - \cos\theta)$,

$$\hat{p} = \frac{\hbar}{i} \frac{\partial}{\partial x},$$

$$\hat{\mathbf{p}} = \frac{\hbar}{i} \nabla,$$

$$[\hat{x}_i, \hat{p}_j] = i\hbar\delta_{ij},$$

$$[\hat{p}_i, f(\hat{\mathbf{x}})] = \frac{\hbar}{i} \frac{\partial f}{\partial x_i}.$$

The equations above illustrate Photo-Electron Energy and interaction free measurements.

- (3) Schrödinger-equation:

$$i\hbar \frac{\partial \psi}{\partial t}(\mathbf{x}, t) = \left(\frac{-\hbar^2}{2m} \nabla^2 + V(\mathbf{x}, t) \right) \psi(\mathbf{x}, t),$$

$$\frac{\partial}{\partial t} \rho(\mathbf{x}, t) + \nabla \cdot \mathbf{J}(\mathbf{x}, t) = 0,$$

$$\rho(\mathbf{x}, t) = |\Psi(\mathbf{x}, t)|^2,$$

$$\mathbf{J}(\mathbf{x}, t) = \frac{\hbar}{m} \text{Im} [\psi^* \nabla \psi].$$

- (4) Fourier transforms: $\psi(x) = \frac{1}{\sqrt{2\pi}} \int dk \phi(k) e^{ikx}$,

$$\Phi(k) = \frac{1}{\sqrt{2\pi}} \int dx \Psi(x) e^{-ikx}, \quad \int dx |\Psi(x)|^2 = \int dk |\Phi(k)|^2. \quad \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{ikx} dx = \delta(k), \quad \frac{1}{(2\pi)^3} \int e^{ik \cdot x} d^3x = \delta^{(3)}(k).$$

From the distribution-integral $\rightarrow$

$$\int_{-\infty}^{\infty} e^{(-ax^2+bx)} dx = \sqrt{\frac{\pi}{a}} e^{\frac{b^2}{4a}}, \quad \text{when } \Re(a) > 0. \quad (4.1)$$

- (5) expectation value:

$$\langle \hat{Q} \rangle \equiv \langle \psi | \hat{Q} | \psi \rangle,$$

$$i\hbar \frac{d}{dt} \langle \hat{Q} \rangle = \langle [\hat{Q}, \mathbf{H}] \rangle, \quad \hat{Q} \text{ time independent.}$$

- (6) Uncertainty:

Consider Schwarz Inequality. Since $[\sigma_Q^2, \sigma_R^2] \geq (\frac{1}{2i} \langle [\hat{Q}, \hat{R}] \rangle)^2$,

$$[\sigma_x^2, \sigma_p^2] = (\frac{1}{2i} \langle i\hbar \rangle)^2 \implies (\frac{i\hbar}{2i})^2 \quad (4.2)$$

Resulting to $[\frac{\hbar^2}{4}]$

$$\text{So, } [\sigma_x^2, \sigma_p^2] = \frac{\hbar^2}{2}$$

$$\Delta A \equiv |A - \langle \hat{A} \rangle \mathbf{1} \psi|$$

$$(\Delta A)^2 = \langle \hat{A}^2 \rangle - \langle \hat{A} \rangle^2 \geq 0.$$

$$\Delta A \Delta B \geq \langle \psi | \frac{1}{2i} [\hat{A}, \hat{B}] | \psi \rangle$$

- (7) saturation: $(\hat{B} - \langle \hat{B} \rangle \mathbf{1}) | \psi \rangle = i\lambda(\hat{A} - \langle \hat{A} \rangle \mathbf{1}) | \psi \rangle, \lambda \in \mathbb{R}.$

$$\Delta x \Delta p \geq \frac{\hbar}{2}. \quad \text{For } \psi = \exp(-\frac{1}{4} \frac{x^2}{\Delta^2}),$$

$$\Delta x = \Delta \text{ and } \Delta p = \frac{\hbar}{2\Delta}.$$

$$\Delta H \Delta t \geq \frac{\hbar}{2}, \Delta t \equiv \frac{\Delta Q}{|\frac{dQ}{dt}|}.$$

- (8) stationary state: $\psi(\mathbf{x}, t) = \psi(\mathbf{x}) e^{-\frac{iEt}{\hbar}}$

$$-\frac{\hbar^2}{2m} \nabla^2 \psi(\mathbf{x}) + \mathbf{V}(\mathbf{x}) \psi(\mathbf{x}) = E \psi(\mathbf{x}) \quad (4.3)$$

- (9) variational-principle: $E_{gs} \leq \langle \psi, \hat{H} \psi \rangle$, for all normalized ψ .

- (10) Hellmann-Feynmann: $\hat{H}(\lambda) \psi_n(\lambda) = E_n(\lambda) \psi_n(\lambda) \longrightarrow \frac{dE_n(\lambda)}{d\lambda} = \langle \psi_n(\lambda), \frac{d\hat{H}}{d\lambda} \psi_n(\lambda) \rangle$.

- (11) virial Theorem: One dimension/Single particle moving under a central potential:

$$\langle \frac{\hat{p}^2}{2m} \rangle = \frac{1}{2} \langle \hat{x} \frac{dV}{dx} \rangle \quad (4.4)$$

- (12) Central potential: $\langle \frac{\hat{p}^2}{2m} \rangle = \frac{1}{2} \langle r \frac{\partial V}{\partial r} \rangle$.

- (13) Commutator Identities:

$$[A, BC] = [A, B]C + B[A, C]$$

$$e^A B e^{-A} = e^{ad_A} B = B + [A, B] + \frac{1}{2!} [A, [A, B]] + \frac{1}{3!} [A, [A, [A, B]]] + \dots, \quad (4.5)$$

This is Hadamards' Lemma using Taylors' expansion, with

$$(ad_A) \text{ as an adjoint action; } (ad_A(X) = [A, X]) \quad (4.6)$$

Note that $e^A e^B \neq e^{A+B} = e^C$, with $C = A + B + \frac{1}{2}[A, B] + \frac{1}{12}([A, [A, B]] - \frac{1}{12}[B, [A, B]]) + \dots$, Its' not evident what the higher terms(...) refer to, although its' been expanded and expressed in terms of commutators.

$$e^A B e^{-A} = B + [A, B], \text{ if } [A, [A, B]] = 0,$$

$$[B, e^A] = [B, A] e^A, \text{ if } [A, [A, B]] = 0,$$

$$e^{A+B} = e^A e^B e^{-\frac{1}{2}[A, B]} = e^B e^A e^{\frac{1}{2}[A, B]} \text{ if } [[A, B], A] = [[A, B], B] = 0.$$

- (14) matrix exponential: $e^{iM\theta} = \mathbf{1} \cos \theta + i \mathbf{M} \sin \theta$,

- (15) index-manipulation:

$$\mathbf{a} \cdot \mathbf{b} = a_i b_i,$$

$$\delta_{ij} B_j = B_i, \delta_{ii} = 3,$$

$$(\mathbf{a} \times \mathbf{b})_i = \epsilon_{ijk} a_j b_k, \epsilon_{123} = 1$$

$$\epsilon_{ijk} \epsilon_{ipq} = \delta_{jp} \delta_{kq} - \delta_{jp} \delta_{kp},$$

$$\epsilon_{ijk} \epsilon_{ijq} = 2\delta_{kq} \quad (4.7)$$

- (16) linear algebra: matrix representation in the basis $(v_1, \dots, v_n)$: $T v_i = \sum_k T_{kj} v_k$.

- (17) basis change:

$$u_k = \sum_j A_{jk} v_j,$$

$$T(u) = A^{-1} T(v) A \quad (4.8)$$

- (18) Inner-product:

$$\langle v, v \rangle \geq 0, \text{ for all } v, \langle v, v \rangle = 0 \iff v = 0, \langle u, v \rangle = \langle vu \rangle^* \quad (4.9)$$

- (19) Schwarz inequality:

$$| \langle u, v \rangle | \leq \|u\| \|v\| \quad (4.10)$$

- (20) Complex-vector-space:

$$\langle v, Tv \rangle = 0,$$

$$\forall v \in V \longrightarrow T = 0.$$

- (21) adjoint:

$$\langle u, Tv \rangle = \langle T^\dagger u, v \rangle,$$

$$(T^\dagger)^\dagger = T,$$

$$(ST)^\dagger = T^\dagger S^\dagger.$$

- (22) Hermitian operator T: $T^\dagger = T$

- (23) unitary operator U: $U^\dagger U = UU^\dagger = \mathbf{1}$.

- (24) orthogonal-projector: $P_u: V \longrightarrow U$,

$$P_u P_u = P_u,$$

$$P_u^\dagger = P_u,$$

$$V = \text{range } P_u \oplus \text{null } P_u.$$

- (25) bra-kets:

$$\langle u | v \rangle \equiv \langle u, v \rangle,$$

$$| Tv \rangle \equiv T | v \rangle,$$

$$\langle Tv | = \langle v | T^\dagger,$$

$$\langle u | T^\dagger v \rangle = \langle Tu | v \rangle.$$

$$| \alpha_1 v_1 + \alpha_2 v_2 \rangle = \alpha_1 | v_1 \rangle + \alpha_2 | v_2 \rangle,$$

$$\langle \alpha_1 v_1 + \alpha_2 v_2 | = \alpha_1^* \langle v_1 | + \alpha_2^* \langle v_2 |.$$

$$\mathbf{T} = |u\rangle \langle w| \longrightarrow T^\dagger = |w\rangle \langle u|,$$

$$\text{tr} \mathbf{T} = \langle \mathbf{w} | \mathbf{u} \rangle.$$

- (26) orthonormal-basis $|i\rangle$:

$$\langle i | j \rangle = \delta_{ij},$$

$$\mathbf{1} = \sum_i |i\rangle \langle i|.$$

$$T_{ij} = \langle i | T | j \rangle \longleftrightarrow \mathbf{T} = \sum_{i,j} T_{i,j} |i\rangle \langle j|,$$

$$\langle i | T^\dagger | j \rangle^*,$$

$$(T^\dagger)_{ij} = (T_{ji})^*.$$

$$[M, M^\dagger] = 0 \longleftrightarrow \mathbf{M} \text{ is unitarily diagonalizable.}$$

- (27) Spectral-Theorem:

$$T^\dagger = \mathbf{T} \longrightarrow \mathbf{T} = \sum_k \lambda_k P_k,$$

$$P_k^\dagger = P_k,$$

$$P_k P_l = \delta_{kl} P_l,$$

$$\sum_k P_k = \mathbf{1}.$$

- (28) Schrödinger picture:

$$|\psi, t\rangle = U(t, t_0) |\psi, t_0\rangle,$$

$$\hat{H}(t) = i\hbar \frac{\partial \mathcal{U}(t, t_0)}{\partial t} \mathcal{U}(t_0, t),$$

$$U(t, 0) = \exp\left(-\frac{i\hat{H}t}{\hbar}\right), \text{ for } \hat{H} \text{ time independent.}$$

- (29) Heisenberg picture:

$$\hat{A}_H(t) \equiv U^\dagger(t, 0) \hat{A}_S U(t, 0),$$

$$i\hbar \frac{d\hat{A}_H(t)}{dt} = [\hat{A}_H(t)],$$

$$\hat{H}_H(t) + i\hbar \frac{\partial \hat{A}_H(t)}{\partial t}.$$

- (30) harmonic oscillator:

$$\hat{H} = \frac{1}{2m} \hat{p}^2 + \frac{1}{2} m \omega^2 \hat{x}^2 = \hbar \omega \left(\hat{N} + \frac{1}{2} \right),$$

$$\hat{N} = \hat{a}^\dagger \hat{a},$$

$$\hat{a} = \frac{1}{\sqrt{2}L_0} \left(\hat{x} + \frac{i\hat{p}}{m\omega} \right),$$

$$\hat{a}^\dagger = \frac{1}{\sqrt{2}L_0} \left(\hat{x} - \frac{i\hat{p}}{m\omega} \right), L_0^2 = \frac{\hbar}{m\omega},$$

$$\hat{x} = \frac{L_0}{\sqrt{2}} (\hat{a} + \hat{a}^\dagger),$$

$$\hat{p} = \frac{i}{\sqrt{2}} \frac{\hbar}{L_0} (\hat{a}^\dagger - \hat{a}),$$

$$[\hat{a}, \hat{a}^\dagger] = 1,$$

$$[\hat{N}, \hat{a}] = -\hat{a},$$

$$\hat{a}\varphi_0 = 0,$$

$$\varphi_0(x) = N_0 \exp\left(-\frac{1}{2} \frac{x^2}{L_0^2}\right),$$

$$N_0^2 = \frac{1}{\sqrt{\pi}L_0},$$

$$\varphi_n = \frac{1}{\sqrt{n!}} (a^\dagger)^n \varphi_0,$$

$$\hat{H}\varphi_n = \hbar\omega\left(n + \frac{1}{2}\right)\varphi_n,$$

$$\hat{N}\varphi_n = n\varphi_n,$$

$$\langle \varphi_m, \varphi_n \rangle = \delta_{mn},$$

$$\hat{a}^\dagger \varphi_n = \sqrt{n+1} \varphi_{n+1},$$

$$\hat{a} \varphi_n = \sqrt{n} \varphi_{n-1},$$

$$\hat{x}_H(t) = \hat{x} \cos \omega t + \frac{\hat{p}}{m\omega} \sin \omega t,$$

$$\hat{p}_H(t) = \hat{p} \cos \omega t - m\omega \hat{x} \sin \omega t.$$

• (31) coherent-states:

$$| \alpha \rangle \equiv e^{\alpha \hat{a}^\dagger - \alpha^* \hat{a}} | 0 \rangle,$$

$$\alpha = \frac{\langle \hat{x} \rangle}{\sqrt{2}L_0} + i \frac{\langle \hat{p} \rangle L_0}{\sqrt{2}\hbar} \in \mathbb{C} \quad (4.11)$$

$$\hat{a} | \alpha \rangle = \alpha | \alpha \rangle,$$

$$| \alpha \rangle = e^{-\frac{1}{2}|\alpha|^2} e^{\alpha \hat{a}^\dagger} | 0 \rangle,$$

$$| \alpha, t \rangle = e^{-i\omega \frac{t}{2}} | e^{-i\omega t} \alpha \rangle.$$

- See (4.3.1) for the quantization of $\hat{L}_z$, and $\hat{L}^2$ for angular momentum Eigenvalues. (32) orbital-angular-momentum

$$\hat{L}_x = \hat{y}\hat{p}_z - \hat{z}\hat{p}_y, \quad \hat{L}_y = \hat{z}\hat{p}_x - \hat{x}\hat{p}_z, \quad \hat{L}_z = \hat{x}\hat{p}_y - \hat{y}\hat{p}_x \quad (4.12)$$

$$[\hat{L}_x, \hat{L}_y] = i\hbar\hat{L}_z,$$

$$[\hat{L}_y, \hat{L}_z] = i\hbar\hat{L}_x,$$

$$[\hat{L}_z, \hat{L}_x] = i\hbar\hat{L}_y,$$

$$\hat{L}^2 \equiv \hat{L}_x\hat{L}_x + \hat{L}_y\hat{L}_y + \hat{L}_z\hat{L}_z, [\hat{L}^2, \hat{L}_i] = 0.$$

$$\nabla^2 = \frac{1}{r} \frac{\partial^2}{\partial r^2} r - \frac{1}{r^2} \frac{\hat{L}^2}{\hbar^2},$$

$$\hat{L}^2 = -\hbar^2 \left(\frac{\partial^2}{\partial \theta^2} + \cot \theta \frac{\partial}{\partial \theta} + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right),$$

$$\hat{L}_z = \frac{\hbar}{i} \frac{\partial}{\partial \theta},$$

$$\hat{L}_{\pm} = \hbar e^{\pm i\phi} \left(\pm \frac{\partial}{\partial \theta} + i \cot \theta \frac{\partial}{\partial \theta} \right).$$

$$Y_{\ell m}(\theta, \phi) = \langle \theta \phi | \ell m \rangle = Y_{\ell m}(\Omega) \equiv \mathcal{N}_{\ell m} P_{\ell}^m(\cos \theta) e^{im\theta}.$$

$$L_z Y_{\ell m} = \hbar m Y_{\ell m},$$

$$L^2 Y_{\ell m} = \hbar^2 \ell(\ell+1) Y_{\ell m},$$

$$\int d\Omega Y_{\ell' m'}^*(\Omega) = \delta_{\ell' \ell} \delta_{m' m},$$

$$\sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} Y_{\ell m}^*(\Omega) = \delta(\cos \theta - \cos \theta') \delta(\phi - \phi').$$

$$Y_{0,0}(\theta, \phi) = \frac{1}{\sqrt{4\pi}},$$

$$Y_{1,\pm 1}(\theta, \phi) = \mp \sqrt{\frac{3}{8\pi}} \sin \theta e^{\pm i\phi},$$

$$Y_{1,0}(\theta, \phi) = \sqrt{\frac{3}{4\pi}} \cos \theta.$$

- (33) central potentials:

$$\psi(r, \theta, \phi) = \frac{u(r)}{r} Y_{\ell m}(\theta, \phi),$$

$$\left(-\frac{\hbar^2}{2m} \frac{d^2}{dr^2} + V(r) + \frac{\hbar^2 \ell(\ell+1)}{2mr^2} \right) u(r) = u(r), u(r) \approx r^{\ell+1}, \text{ as } r \rightarrow 0.$$

- (34) Spin one-half

$$\hat{H} = -\mu \cdot B, \hat{\mu} = g \frac{e\hbar}{2mc} \frac{1}{\hbar} \hat{S},$$

$$\mu_B = \frac{e\hbar}{2m_e C}, \mu_e = -2 \mu_B \frac{\hat{S}}{\hbar}.$$

$$|1\rangle \equiv |Z; +\rangle = |+\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix},$$

$$|2\rangle \equiv |Z; -\rangle = |-\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix},$$

$$\hat{S}_i = \frac{\hbar}{2}\sigma_i, \sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

$$\sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$[\sigma_i, \sigma_j] = 2i\epsilon_{ijk}\sigma_k,$$

$$[\hat{S}_i, \hat{S}_j] = i\hbar\epsilon_{ijk}\hat{S}_k,$$

$$\sigma_i\sigma_j = \delta_{ij}\mathbf{1} + i\epsilon_{ijk}\sigma_k, (\sigma \cdot \mathbf{a})(\sigma \cdot \mathbf{b}) = \mathbf{a} \cdot \mathbf{b}\mathbf{1} + i\sigma \cdot (\mathbf{a} \times \mathbf{b}),$$

$$e^{i\mathbf{a} \cdot \boldsymbol{\sigma}} = \mathbf{1} \cos a + i\sigma \cdot \hat{\mathbf{a}} \sin a,$$

$$\mathbf{a} = \hat{\mathbf{a}}a, a = |\mathbf{a}|.$$

$$\hat{S}_n \equiv \mathbf{n} \cdot \hat{\mathbf{S}} = \frac{\hbar}{2}\mathbf{n} \cdot \boldsymbol{\sigma}, \hat{S}_n | \mathbf{n}; \pm \rangle = \pm \frac{\hbar}{2} | \mathbf{n}; \pm \rangle,$$

$$| \mathbf{n} \rangle \equiv | \mathbf{n}; + \rangle,$$

$$\langle \mathbf{n} | \hat{\mathbf{S}} | \mathbf{n} \rangle = \frac{\hbar}{2}\mathbf{n},$$

$$| \mathbf{n}; + \rangle = \cos \frac{\theta}{2} | + \rangle + \sin \frac{\theta}{2} \mathbf{e}^{i\theta} | - \rangle,$$

$$| \mathbf{n}; - \rangle = -\sin \frac{\theta}{2} \mathbf{e}^{-i\theta} | + \rangle + \cos \frac{\theta}{2} | - \rangle,$$

$$\hat{R}_n(\alpha) \equiv \exp(-\frac{i}{\hbar}\alpha\hat{S}_n),$$

$$\hat{R}_n(\alpha) | \mathbf{n}' \rangle = | \mathbf{n}'' \rangle, \text{ with } \mathbf{n}'' = \mathcal{R}_n(\alpha)\mathbf{n}'$$

$$\hat{R}_n(\alpha)\hat{\mathbf{S}}\hat{R}_n(\alpha) = \mathcal{R}(\alpha)\hat{\mathbf{S}}$$

• (35) Spin precession: $\hat{H} = \omega_L \cdot \hat{\mathbf{S}},$

$$\omega_L = -\gamma\mathbf{B}, \hat{\mu} = \gamma\hat{\mathbf{S}}.$$

• (36) general angular momentum:

$$[\hat{J}_i, \hat{J}_j] = i\hbar\epsilon_{ijk}\hat{J}_k \longleftrightarrow \hat{\mathbf{J}} \times \hat{\mathbf{J}} = i\hbar\hat{\mathbf{J}},$$

$$[\hat{\mathbf{J}}^2, \hat{J}_i] = 0.$$

$$\hat{\mathbf{J}}_{\pm} = \hat{\mathbf{J}}_x \pm i\hat{\mathbf{J}}_y, (\hat{J}_{\pm})^{\dagger} = \hat{J}_{\mp},$$

$$\hat{\mathbf{J}}_x = \frac{1}{2}(\hat{\mathbf{J}}_+ + \hat{\mathbf{J}}_-), \hat{\mathbf{J}}_y = \frac{1}{2i}(\hat{\mathbf{J}}_+ - \hat{\mathbf{J}}_-),$$

$$[\hat{\mathbf{J}}_z, \hat{\mathbf{J}}_{\pm}] = \pm \hbar \hat{\mathbf{J}}_{\pm}, [\hat{\mathbf{J}}_+, \hat{\mathbf{J}}_-] = 2\hbar \hat{\mathbf{J}}_z,$$

$$[\hat{\mathbf{J}}^2, \hat{\mathbf{J}}_{\pm}] = 0,$$

$$\hat{\mathbf{J}}^2 = \hat{\mathbf{J}}_+ \hat{\mathbf{J}}_- + \hat{\mathbf{J}}_z^2 - \hbar \hat{\mathbf{J}}_z = \hat{\mathbf{J}}_- \hat{\mathbf{J}}_+ + \hat{\mathbf{J}}_z^2 + \hbar \hat{\mathbf{J}}_z.$$

$$\hat{\mathbf{J}}^2 |\mathbf{j}\mathbf{m}\rangle = \hbar^2 \mathbf{j}(\mathbf{j} + 1) |\mathbf{j}\mathbf{m}\rangle = \hbar \mathbf{m} |\mathbf{j}\mathbf{m}\rangle,$$

$$\mathbf{m} = -\mathbf{j}, \dots, \mathbf{j}$$

$$\hat{\mathbf{J}}_{\pm} |\mathbf{j}\mathbf{m}\rangle = \hbar \sqrt{\mathbf{j}(\mathbf{j} + 1) - \mathbf{m}(\mathbf{m} \pm 1)} |\mathbf{j}, \mathbf{m} \pm 1\rangle.$$

$$\hat{R}_n^\dagger(\alpha) \hat{\mathbf{J}} \hat{R}_n(\alpha) = \mathcal{R}_n(\alpha) \hat{\mathbf{J}},$$

$$\hat{\mathbf{R}}_n(\alpha) = e^{-i \frac{\alpha}{\hbar} \mathbf{n} \cdot \mathbf{J}}.$$

• (37) addition of angular momentum:

$$\hat{\mathbf{J}} = \hat{\mathbf{J}}_1 + \hat{\mathbf{J}}_2.$$

• (38) uncoupled basis: $|j_1 j_2; \mathbf{m}_1 \mathbf{m}_2\rangle$ CSCO: $(\hat{\mathbf{J}}_1^2, \hat{\mathbf{J}}_2^2, \hat{J}_{1z}, \hat{J}_{2z})$

• (39) coupled basis:

$|j_1 j_2; \mathbf{j}\mathbf{m}\rangle$ CSCO:

$$(\hat{\mathbf{J}}_1^2, \hat{J}_2^2, \hat{\mathbf{j}}^2, \hat{J}_z).$$

$$j_1 \otimes j_2 = (j_1 + j_2) \oplus (j_1 + j_2 - 1) \oplus \dots \oplus |j_1 - j_2\rangle.$$

$$|\mathbf{j}_1 \mathbf{j}_2; \mathbf{j}\mathbf{m}\rangle = \sum_{\mathbf{m}_1 + \mathbf{m}_2 = \mathbf{m}} |\mathbf{j}_1 \mathbf{j}_2; \mathbf{m}_1 \mathbf{m}_2\rangle \underbrace{\langle \mathbf{j}_1 \mathbf{j}_2; \mathbf{m}_1 \mathbf{m}_2 | \mathbf{j}_1 \mathbf{j}_2; \mathbf{j}\mathbf{m}\rangle}_{\text{Clebsch-Gordan-coefficient}} \quad (4.13)$$

$$\hat{\mathbf{J}}_1 \cdot \hat{\mathbf{J}}_2 = \frac{1}{2} (\hat{J}_{1+} \hat{J}_{2-} + \hat{J}_{1-} \hat{J}_{2+}) + \hat{J}_{1z} \hat{J}_{2z} = \frac{1}{2} (\hat{\mathbf{J}}^2 - \hat{\mathbf{J}}_1^2 - \hat{\mathbf{J}}_2^2).$$

• (40) Hydrogen-atom:

$$\hat{H} = \frac{\hat{\mathbf{p}}^2}{2\mathbf{m}} - \frac{Ze^2}{r}, Z = 1 \text{ for hydrogen.}$$

$$E_n = -\frac{Z^2 e^2}{2a_0} \frac{1}{\mathbf{n}^2}, a_0 = \frac{\hbar^2}{me^2} \simeq 52.9 \text{ \AA},$$

$$\frac{e^2}{2a_0} = \frac{1}{2} \mathbf{m} \mathbf{c}^2 \alpha^2 \equiv \text{Ry} \simeq 13.6 \text{ eV.}$$

$$\psi_{n\ell m}(\mathbf{X}) = \mathcal{N} \left(\frac{r}{a_0} \right)^\ell (\text{polynomial} \in \frac{r}{a_0} \text{ of degree } [\mathbf{n} - (\ell + 1)]) e^{-\frac{Zr}{\mathbf{n}a_0}} \mathbf{Y}_{\ell m}(\theta, \phi),$$

$$\psi_{100}(\mathbf{r}, \theta, \phi) = \sqrt{\frac{Z^3}{\pi a_0^3}} e^{-\frac{Z}{a_0}}$$

$$Z = 1: \langle r \rangle = \frac{1}{2} a_0 (3n^2 - \ell(\ell + 1)), \langle \frac{1}{r} \rangle = \frac{1}{a_0 n^2},$$

$$\langle \frac{1}{r^2} \rangle = \frac{1}{a_0^2 n^3 (\ell + \frac{1}{2})},$$

$$\langle \frac{1}{r^3} \rangle = \frac{1}{a_0^3 n^3 \ell (\ell + \frac{1}{2}) (\ell + 1)}.$$

- (41) Fine structure corrections of the Hydrogen atom:

$$E_{n\ell j m_j} = -\frac{e^2}{2a_0} \frac{1}{n^2} \left(1 + \frac{\alpha^2}{n^2} \left[\frac{\mathbf{n}}{j + \frac{1}{2}} - \frac{3}{4} \right] \right).$$

- (42) Density matrix:

$$E = [(\mathbf{p}_1, |\psi_1\rangle), \dots, (p_n, |\psi_n\rangle)], p_1, \dots, p_n > 0,$$

$$p_1 + \dots + p_n = 1,$$

$$\rho_E \equiv \sum_{a=1}^n p_a |\psi_a\rangle \langle \psi_a|, \langle \hat{Q} \rangle_E = \text{tr}(Q \rho_E). \text{ General } \rho \text{ is positive semi-definite, and } \text{tr}(\rho) = 1.$$

$$\text{Pure state} \longleftrightarrow \text{tr}(\rho^2) = 1.$$

$$\text{Spin-one-half-density-matrix: } \rho = \frac{1}{2}(\mathbf{1} + \mathbf{a} \cdot \boldsymbol{\sigma}),$$

$$|\mathbf{a}| \leq 1.$$

- (43) Time evolution:

$$i\hbar \frac{\partial \rho}{\partial t} = [\hat{H}, \rho].$$

- (44) Schmidt decomposition:

$$|\psi_{AB}\rangle = \sum_{k=1}^r \sqrt{p_k} |\mathbf{k}_A\rangle \otimes |k_B\rangle,$$

$$r \leq \mathbf{d}_A \leq \mathbf{d}_B,$$

$$\rho_A = \sum_{k=1}^r p_k |\mathbf{k}_A\rangle \langle \mathbf{k}_A|,$$

$$\rho_B = \sum_{k=1}^r p_k |k_B\rangle \langle \mathbf{k}_B|,$$

$$\langle \mathbf{k}_A | \mathbf{k}'_A \rangle = \delta_{\mathbf{k}, \mathbf{k}'},$$

$$\langle k_B | k'_B \rangle = \delta_{k, k'}$$

- (45) Lindblad equation:

$$\frac{\partial \rho}{\partial t} = \frac{1}{i\hbar} [\mathbf{H}, \rho] + \sum_k (\mathbf{L}_k \rho \mathbf{L}_k^\dagger - \frac{1}{2} [\mathbf{L}_k^\dagger \mathbf{L}_k, \rho]) \quad (4.14)$$

- (46) electromagnetic couplings:

$$\hat{H} = \frac{1}{2m} (\hat{\mathbf{p}})^2 - \frac{q}{c} \mathbf{A}(\hat{\mathbf{x}}, t)^2 + q \phi(\hat{\mathbf{x}}, t) \text{ (no spin).}$$

- (47) Go to equation (1.7) Gauge-transformations:

$$\mathbf{A}' = \mathbf{A} + \nabla \Lambda,$$

$$\phi' = \phi - \frac{1}{c} \frac{\partial \Lambda}{\partial t},$$

$$\Psi' = \exp(i \frac{q \Lambda}{\hbar c}) \Psi.$$

- (48) Pauli Hamiltonian (electron):

$$\hat{H}_{Pauli} = \frac{1}{2m_e} (\hat{\mathbf{p}} + \frac{e}{c} \mathbf{A})^2 + \frac{e \hbar}{2m_e c} \boldsymbol{\sigma} \cdot \mathbf{B} - e \phi(\hat{\mathbf{x}}, t).$$

- (49) Time-independent Perturbation: non-degenerate:

$$|n\rangle_\lambda = |n^{(0)}\rangle - \lambda \sum_{k \neq n} \frac{\delta \mathbf{H}_{kn}}{\mathbf{E}_k^{(0)} - \mathbf{E}_n^{(0)}} |k^{(0)}\rangle + \mathcal{O}(\lambda^2),$$

$$E_n(\lambda) = E_n^{(0)} + \lambda \delta \mathbf{H}_{nn} - \lambda^2 \sum_{k \neq n} \frac{|\delta \mathbf{H}_{kn}|^2}{\mathbf{E}_k^{(0)} - \mathbf{E}_n^{(0)}} + \mathcal{O}(\lambda^3).$$

- (50) Degeneracy lifted at $\mathcal{O}(\lambda)$: good-basis:

$$\delta \mathbf{H}_{\mathcal{I}} \equiv \langle \Psi_{\mathcal{I}}^{(0)} | \delta \mathbf{H} | \Psi_{\mathcal{J}}^{(0)} \rangle = \mathbf{E}_{\mathcal{I}\mathcal{J}}^{(1)} \delta_{\mathcal{I}\mathcal{J}},$$

$$|\Psi_{\mathcal{I}}\rangle_\lambda = |\Psi_{\mathcal{I}}^{(0)}\rangle - \lambda \sum_p \frac{\delta \mathbf{H}_{p\mathcal{I}}}{\mathbf{E}_p^{(0)} - \mathbf{E}_n^{(0)}} |\mathbf{p}^{(0)}\rangle + \sum_{K \neq \mathcal{I}} \frac{|\Psi_K^{(0)}\rangle}{\mathbf{E}_{n\mathcal{I}}^{(1)} - \mathbf{E}_{nK}^{(0)}} \sum_p \frac{\delta \mathbf{H}_{Kp} \delta \mathbf{H}_{p\mathcal{I}}}{\mathbf{E}_{(0)p} - \mathbf{E}_{(0)n}} + \mathcal{O}(\lambda^2),$$

$$\mathbf{E}_{n\mathcal{I}}(\lambda) = \mathbf{E}_n^{(0)} + \lambda \delta \mathbf{H}_{\mathcal{I}\mathcal{I}} - \lambda^2 \sum_p \frac{|\delta \mathbf{H}_{p\mathcal{I}}|^2}{\mathbf{E}_p^{(0)} - \mathbf{E}_n^{(0)}} + \mathcal{O}(\lambda^3).$$

- (51) WKB-quantization with (a, b) turning-points or hard walls:

$$\int_a^b k(x') dx = (\mathbf{n} + \beta) \pi,$$

$$\mathbf{n} = 0, 1, 2, \dots,$$

$$\beta = 1 - \frac{1}{4} \text{ (Number of turning points).}$$

- (52) Time-dependent perturbations:

$$\hat{\mathbf{H}}(t) = \hat{\mathbf{H}}^{(0)} + \delta \mathbf{H}(t), |\Psi(t)\rangle \equiv e^{i \hat{\mathbf{H}}^{(0)} \frac{t}{\hbar}} |\Psi(t)\rangle,$$

$$i \hbar \frac{d}{dt} |\tilde{\Psi}(t)\rangle = \tilde{\delta \mathbf{H}}(t) |\Psi(t)\rangle,$$

$$\delta \mathbf{H}(t) \equiv e^{i \hat{\mathbf{H}}^{(0)} \frac{t}{\hbar}} \delta \mathbf{H}(t) e^{-i \hat{\mathbf{H}}^{(0)} \frac{t}{\hbar}},$$

$$|\tilde{\Psi}(t)\rangle = |\Psi(0)\rangle + \int_0^t \frac{\delta \mathbf{H}(t')}{i \hbar} |\Psi(0)\rangle dt' + \mathcal{O}(\delta \mathbf{H}^2)$$

- (53) Fermis'-golden-rule:

$$\hat{\mathbf{H}} = \hat{\mathbf{H}}^{(0)} + \mathbf{V},$$

$$\hat{\mathbf{H}} = \hat{\mathbf{H}}^{(0)} + 2\mathbf{H}' \cos \omega t,$$

$$w = \frac{2\pi}{\hbar} |\mathbf{V}_f|^2 \rho(\mathbf{E}_f),$$

$$w = \frac{2\pi}{\hbar} |\mathbf{H}'_f| 2\rho(\mathbf{E}_f)$$

- (54) Adiabatic-approximation:

$$|\Psi(t)\rangle \simeq e^{i\theta_k(t)} e^{i\gamma_k(t)} |\Psi_k(t)\rangle,$$

$$\hat{\mathbf{H}}(t) |\Psi_k(t)\rangle = \mathbf{E}_k(t) |\Psi_k(t)\rangle,$$

$$\theta_k(t) = -\frac{1}{\hbar} \int_0^t \mathbf{E}_k(\mathbf{t}') d\mathbf{t}',$$

$$\gamma_k(t) = \int_0^t \nu_k(t') d\mathbf{t}',$$

$$\nu_k(t) = i\langle \Psi_k(\mathbf{t}) | \dot{\Psi}_k(t) \rangle.$$

- (55) Berrys' phase:

$$\gamma_n(\Gamma_{if}) = \int_{\Gamma_{if}} \mathbf{A}_n(\mathbf{R}) \cdot d\mathbf{R}, \mathbf{A}_n(\mathbf{R}) \equiv i\langle \Psi_n(\mathbf{R}) | \nabla_{\mathbf{R}} | \Psi_n(\mathbf{R}) \rangle.$$

- (56) Scattering-on-the-half-line:

$$\Psi(\mathbf{x}) = e^{i\delta} \sin(\mathbf{k}\mathbf{x} + \delta),$$

$$\Psi_s(\mathbf{x}) = e^{i\delta} \sin \delta e^{ikx},$$

$$\mathbf{x} > \mathbf{R}$$

- (57) Time-delay:

$$\Delta t = 2\hbar\delta'(\mathbf{E}_0).$$

- (58) Levinsons' Theorem:

$$N_b = \frac{1}{\pi} ((\delta)(0) - \delta(\infty)).$$

- (59) Resonance, Breit-Wigner:

$$|\Psi_s|^2 \simeq \frac{\frac{1}{4}\Gamma^2}{(\mathbf{E}-\mathbf{E}_\alpha)^2 + \frac{1}{4}\Gamma^2}.$$

Energy dependence of the squared scattering amplitude $|\Psi_s|^2$ near resonance.

• (60) Scattering in 3D:

$$\Psi(\mathbf{r}) = \varphi(\mathbf{r}) + \Psi_s(\mathbf{r}) \simeq e^{ikz} + f_k(\theta, \phi) \frac{e^{ikr}}{r}, \mathbf{r} \gg a. \quad (4.15)$$

$$\frac{d\sigma}{d\Omega} = |f_k(\theta, \phi)|^2,$$

$$\sigma = \int |f_k(\theta, \phi)|^2 d\Omega.$$

• (61) Rayleigh:

$$e^{ikz} = \sqrt{4\pi} \sum_{\ell=0}^{\infty} \sqrt{2\ell+1} i^\ell \mathbf{Y}_{\ell 0}(\theta) j_\ell(kr).$$

• (62) Phase shifts:

$$f_k(\theta) = \frac{\sqrt{(4\pi)}}{k} \sum_{l=0}^{\infty} \sqrt{2l+1} \mathbf{Y}_{l0}(\theta) e^{i\delta_l} \sin \delta_l,$$

$$\sigma = \frac{4\pi}{k^2} \sum_{l=0}^{\infty} (2l+1) \sin^2 \delta_l.$$

$$\Psi_{\mathbf{r}}|_l = \mathbf{A}_l \mathbf{j}_l(kr) + \mathbf{B}_l \mathbf{n}_l \mathbf{k} r \mathbf{Y}_{l0}(\theta),$$

$$\mathbf{r} > \mathbf{a},$$

$$\tan \delta_l \equiv - \frac{\mathbf{B}_{(l)}}{\mathbf{A}_{(l)}}.$$

Chapter 5 The Simple Harmonic Oscillator

When Energy is quantized, the only allowed values for $E = \epsilon \hbar \omega$, are values that yield solutions in the physical Hilbert space: $E_n = (n + \frac{1}{2}) \hbar \omega$, $(n) = 0, 1, 2, \dots$

For each value of (n) , the corresponding polynomials is of (n_{th}) order, called Hermite polynomials. We know that: $[x, p] = i \hbar$, and that the Hamiltonian $\hat{H} = \frac{p^2}{2m} + \frac{1}{2} m \omega^2 \hat{x}^2$. The Quantum Harmonic Oscillator is an extension of the Classical Oscillator. We expect that Energy eigenstates, only exist for certain quantized values of (E) . This is a differential equation:

$$-\frac{\hbar^2}{2m} \frac{d^2 \psi(x)}{dx^2} + \frac{1}{2} m \omega^2 x^2 \psi(x) = E \psi(x) \quad (5.1)$$

The length scale $L_0 = \sqrt{\frac{\hbar}{m\omega}}$, to be called the **oscillator size**, is as relevant to the harmonic oscillator as the Bohr radius (a_0), is to the Hydrogen atom. The ground state energy of the quantum oscillator, is $(\frac{\hbar \omega}{2})$, and (L_0) , is precisely the Classical oscillation amplitude of an oscillator with that Energy. A recursion relation of $(\epsilon = 2n + 1)$ must terminate, for us to get an Energy eigenstate. The corresponding physical Energies $E_n = \frac{\hbar \omega}{2} (2n + 1) = \hbar \omega (n + \frac{1}{2})$, $n = 0, 1, 2, \dots$. The spectrum of this result is non-degenerate, and all energy eigenstates of the harmonic oscillator are bound states of a one-dimensional potential. The polynomial solutions are normalized Hermite polynomials and satisfy 2^{nd} order differential equation. We can write the eigenstate as a function of the physical coordinate (x) , from Hermite polynomials. *The Energy eigenstates are real functions of (x) .* We have full orthonormality of the Energy eigenstates $\langle \varphi_n, \varphi \rangle = \delta_{mn}$, from the normalization constant (N_n) , determined from the condition $\langle \varphi_n, \varphi_n \rangle = \int_{-\infty}^{\infty} dx \varphi_n(x) = 1$. $(\hat{x})$ and $(\hat{p})$ are Hermitian operators. All Energy eigenstates satisfy Energy (E) , so $E \geq \frac{1}{2} \hbar \omega$, since $\langle \hat{H} \rangle_\psi = E \hat{p}$, is $\frac{\hbar}{i} \frac{\partial}{\partial x}$ in *Position* space, and $\hat{x}$ is $i \hbar \frac{\partial}{\partial p}$ in *Momentum* space. The normalized (n_{th}) excited state of the simple harmonic oscillator is:

$$\varphi_n \equiv \frac{1}{\sqrt{N!}} \underbrace{\hat{a}^+ \dots \hat{a}^+}_n \psi_0 = \frac{1}{\sqrt{N!}} (\hat{a}^+)^n \psi_0 \quad (5.2)$$

Since $\hat{N} \varphi_n = \varphi_n$, means $(\hat{N})$ counts the number of creation operators acting on the ground state, and its' Energy is given by:

$$E_n = \hbar \omega (n + \frac{1}{2}) \quad (5.3)$$

$\therefore (\Delta x)_n^2 = L_0^2 (n + \frac{1}{2}) \approx L_0 \sqrt{n}$, comparable to the amplitude of the Classical oscillation of a particle with Energy $\hbar \omega n$. Only for the ground state ϕ_0 , does the product of uncertainties saturate the lower bound given by the Heisenberg uncertainty inequality:

$$(\Delta(x), \Delta(p)) \geq \frac{\hbar}{2} \quad (5.4)$$

Chapter 6 Sample Practice

6.1 More Useful Equations

- d'Alemberts' Principle $\longrightarrow \sum_i \langle \mathbf{F}_i(\mathbf{c}(t), t) - \mathbf{m}_i \mathbf{c}_i(t), \mathbf{v}_i \rangle = 0 \forall$ tangent vectors $\mathbf{v}$ at $\mathbf{M}_{\mathbf{c}(t)}$

- n^{th} Taylor Polynomial about $x = x_0$ for f ,

$$p_n(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2!}(x - x_0)^2 + \dots + \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n,$$

$$\sum_{k=0}^{\infty} \frac{(-i * \frac{\theta}{2} * X)^k}{k!}$$

- Airy equation $\longrightarrow \frac{d^2 \psi}{du^2} - u\psi(u) = 0$

- Jacobi identity $\longrightarrow [X, [Y, Z]] + [Y, [X, Z]] + [Z, [X, Y]] = 0$

- Stokes theorem(Line-Integrals)

$$\iint_S \text{curl} \mathbf{F} \cdot d\mathbf{S} = \int_C \mathbf{F} \cdot d\mathbf{r}, \text{ where } C \text{ is a closed curve, and } \mathbf{F} \text{ is a vector field defined on } C.$$

- A symplectic-manifold (N, ω) , is quantizable (for a particular value of $\hbar$), if $\frac{1}{2\pi\hbar} \int_S \omega \in \mathbb{Z}$, for every closed surface S in N .

- Probability-density-function(Gaussian-distribution):

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \quad (6.1)$$

- Poisson bracket of f and g ; Moyal bracket $\longrightarrow [f, g] =$

$$\frac{\partial f}{\partial q_1} \frac{\partial g}{\partial p_1} + \dots + \frac{\partial f}{\partial q_n} \frac{\partial g}{\partial p_n} - \frac{\partial f}{\partial p_1} \frac{\partial g}{\partial q_1} - \dots - \frac{\partial f}{\partial p_n} \frac{\partial g}{\partial q_n}$$

- Laguerre Polynomials $\longrightarrow$

$$f_{n,l}(r) := L_{n-l-1}^{2l+1} \left(\frac{2r}{nr_B} \right), \text{ where } r_B \text{ is the Borh-radius.}$$

- Rydberg formula $\longrightarrow$

$$\frac{1}{\lambda} = R_H \left(\frac{1}{n^2} - \frac{1}{m^2} \right) \quad (6.2)$$

where R_H is the Rydberg constant.

- Rayleigh formula $\longrightarrow$

$$\frac{8\pi\nu^2 kT_{\Delta\nu}}{c^3} \hbar e^{-\left(\frac{\hbar}{k}\right)\left(\frac{\nu}{T}\right)},$$

- Wiens' formula $\longrightarrow$

$\frac{A\nu^3 e^{-b\nu}}{T_{\Delta\nu}}$, where A and b are constants(does not fit experimental facts for small $\frac{\nu}{T}$, although it avoids the paradox of infinite total energy.

- Radon-Nikodym Theorem(derivative of ν w.r.t μ) $\longrightarrow$

$$\nu(E) = \int_E \rho(s) d\mu(s)$$

There exists a measurable function ρ which is the density of ν w.r.t μ

- Schrödinger operator $\longrightarrow$

$$\lim_{X \rightarrow \infty} \frac{\beta([0, X])}{\beta_q([0, X])} = (2\pi\hbar)^N, \text{ where}$$

$\beta([0, X])$ is a continuous measure space of the set of classic Hamiltonian, $\beta_q([0, X])$ is the discrete measure of the number of eigenvalues $< (X)$, and (N) is the number of dimensions in configuration space.

- Schurs' Theorem $\longrightarrow U^*AU = T = [T_{ij}] \in M_n(\mathbb{C}), T_{ij} = 0(i > j)$.

- Angular momentum in the tensor product space $\longrightarrow$

$$\hat{j}_i = \hat{j}_i^{(1)} \otimes \mathbf{1} + \mathbf{1} \otimes \hat{j}_i^{(2)} \text{ satisfy } [\hat{j}_i, \hat{j}_j] = i\hbar\epsilon_{ijk}\hat{j}_k \text{ acting on } V_1 \otimes V_2$$

- Maxwells'-equations $\longrightarrow$

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}$$

$$\nabla \cdot \mathbf{B} = 0$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

$$\nabla \times \mathbf{B} = \mu_0 \left(\mathbf{J} + \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right)$$

- Abelian/Commutative $\longrightarrow$

$$\forall a, b \in G, \quad a \cdot b = b \cdot a \quad (6.3)$$

- Matrix representation of 2-D Hilbert space $\longrightarrow$

$$H^2 = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \mathbf{I},$$

where H is for the Hadamard gate, and $(\mathbf{I})$ is the identity.

- Leibniz-Rule $\longrightarrow$

$$\frac{d}{dt} \int_{-\infty}^{\infty} |\psi(x, t)|^2 dx = \int_{-\infty}^{\infty} \frac{\partial}{\partial t} |\psi(x, t)|^2 dx = 0 \quad (6.4)$$

- Wave function for a particle in a box: $\longrightarrow$

$$\Psi(x, t) = \sum_{n=1}^{\infty} C_n \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi}{a}x\right) e^{-i\left(\frac{n^2\pi^2\hbar}{2ma^2}\right)t} \quad (6.5)$$

- Ground-state for a particle in a box $\longrightarrow \Psi_n(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi}{a}x\right)$

- Gaussian-integral $\longrightarrow$

If $z^2 = ax^2$, and $dz = \sqrt{a}dx$, then $\int_{-\infty}^{\infty} e^{-ax^2} dx = \sqrt{\frac{\pi}{a}}$ [Highly simplified]

- Hookes' Law $\longrightarrow$

$$F = -\frac{dV}{dx} = -\frac{d}{dx} \left(\frac{1}{2} kx^2 \right) = -kx = ma = m \frac{d^2(x)}{dt^2}$$

- Energy-spectrum of the Transmon qubit $\longrightarrow$

$$\omega_j = \left(\omega - \frac{\delta}{2} \right) j + \frac{\delta}{2} j^2$$

- Pure density operator, and state $\longrightarrow$

$$\text{tr} \rho^2 = \text{tr} \rho = 1; \frac{1}{N} < \text{tr} \rho < 1 \quad (6.6)$$

$\longrightarrow$

positive-definiteness, where $\lambda_j \geq 0$ (λ_j are the eigenvalues, and 'N' is the Hilbert space dimension).

- Traceless-Pauli-matrices $\longrightarrow$

$$\rho = \frac{1}{2} \left[\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + x \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + y \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} + z \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \right] = \frac{1}{2} (\mathbf{I} + \vec{r} \cdot \vec{\sigma}), \text{ where } \vec{r} \in \mathbb{R}^3,$$

and $\vec{\sigma} = (x, y, z)$.

6.1.1 Test your comprehension

Every equation has a meaning, and can be logically explained.

6.1.1.0.1 Test Case A

Q1. Time-dependent Schrödinger Equation:

Describes how a quantum system evolves $\longrightarrow$

$$i\hbar \frac{\partial \psi(x, t)}{\partial t} = \left(-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V(x, t) \right) \psi(x, t) = \hat{H} \psi \quad (6.7)$$

- $\psi(x, t)$: The above equation is a wave-function (Probability amplitude).

One-Dimensional Time-independent Schrödinger Equation $\longrightarrow$

$$-\frac{\hbar^2}{2m} \frac{d^2 \psi}{dX^2} + V\psi = \hat{H}\psi$$

Q2. Heisenberg Uncertainty Principle/Minimum uncertainty wave-packet $\longrightarrow$

$$\Delta x \Delta p \geq \frac{\hbar}{2} \text{ (Note that } \left| \frac{i\hbar \langle \psi | \psi \rangle}{2i} \right|^2 = \sqrt{\frac{\hbar^2}{4}} = \frac{\hbar}{2}).$$

- You cannot measure Position and Momentum precisely at the same time
- More precision in one, less precision in the other

Q4. Particle in a box Energies $\longrightarrow$

$$E_n = \frac{n^2 \pi^2 \hbar^2}{2mL^2}, n = 1, 2, 3, \dots$$

- $\hbar$: reduced Plancks' constant ($\frac{h}{2\pi}$) [h is the Plancks constant].

- i: imaginary unit/Lateral-unit

- $\hat{H}$: Hamiltonian (total energy operator)

Q5. Commutation-relation $\longrightarrow [\hat{x}, \hat{p}] = i\hbar$

Significance:

- Leads to uncertainty principle
- Shows position and momentum, are incompatible observables

6.1.1.0.2 Test Case B

Q6. Particle in a box

(a) Wave-function

Boundary conditions:

- $\psi(0) = 0 \implies \mathbf{B} = 0$

- $\psi(\mathbf{L}) = 0 \sin(k\mathbf{L}) = 0$

So:

$$k = \frac{n\pi}{L}$$

Final wave-function:

$$\psi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right)$$

(b) Energy-Level

$$E_n = \frac{n^2 \pi^2 \hbar^2}{2mL^2}$$

(c) Probability

$$\mathbf{P} = \int_0^{\frac{L}{2}} |\psi_1|^2 dx = \frac{2}{L} \int_0^{\frac{L}{2}} \sin^2\left(\frac{\pi x}{L}\right) dx$$

Result:

$$\mathbf{P} = \frac{1}{2}$$

Q7. Commutator and Uncertainty

(a) Show:

$$[\hat{x}, \hat{p}] = i\hbar$$

Apply to function $f(x)$:

$$\hat{x}\hat{p}f = x(-i\hbar f')$$

$$\hat{p}\hat{x}f = -i\hbar \frac{d}{dx}(xf) = -i\hbar(f + xf')$$

Subtract:

$$[\hat{x}, \hat{p}]f = i\hbar f$$

Thus:

$$[\hat{x}, \hat{p}] = i\hbar$$

(b) Uncertainty-Principle

General relation:

$$\Delta \mathbf{A} \Delta \mathbf{B} \geq \frac{1}{2} | \langle [\mathbf{A}, \mathbf{B}] \rangle |$$

Substitute:

$$\Delta x \Delta p \geq \frac{\hbar}{2}$$

Q9. Hydrogen-Atom

(a) Energy-Level

$$E_n = - \frac{13.6}{n^2} \text{ eV}$$

(b) Quantum-number

- n : Energy-Level
- ℓ : orbital-shape
- m : orientation

(c) Why Energy is Negative

- Electron is bound to nucleus
- Energy is required to remove electron

- Bound-states

Wave-function-collapse

- Before measurement, we have superposition
- Measurement leads to a definite state

A collapse = Transition from probability to reality $\mathbb{R}$

6.1.1.0.3 Additional Notes

I.(a) Completeness

A Hilbert space is complete if:

Every Cauchy sequence $\psi_n \subset \mathcal{H}$ converges to a limit in $\mathcal{H}$

Formally:

$$\|\psi_n - \psi_m\| \rightarrow 0 \implies \exists \psi \in \mathcal{H}, \psi_n \rightarrow \psi$$

(b) Extension to Orthonormal basis

Given orthonormal-set e_1 :

- If not complete, there exists $\psi \neq 0$ orthogonal to all e_1
- Add ψ , normalize enlarge set

Zorns' -Lemma argument:

Maximal orthonormal-set = orthonormal basis

(c) Projection is bounded

Projection-operator $\mathbf{P}$:

$$\mathbf{P}^2 = \mathbf{P}, \mathbf{P}^\dagger = \mathbf{P}$$

For any ψ :

$$\|\mathbf{P}\psi\| \leq \|\psi\|$$

Thus:

$$\|\mathbf{P}\| \leq 1. \text{ therefore bounded.}$$

II. Spectral-Theorem

(a) Statement

For self-adjoint-operator $\hat{A}$:

$$\hat{A} = \int \lambda \, dE(\lambda)$$

- $E(\lambda)$: Projection-valued-measure
- Generalizes diagonalization

(b) Spectrum-Types

- Discrete eigenvalues (Bound-states)
 - Continuous scattering-states
- (c) Measurement-Interpretation

Probability of outcome in set Δ :

$$P(\Delta) = \langle \psi | E(\Delta) | \psi \rangle$$

Measurement-outcome = spectrum of operator

III. Operators and Dynamics

Time Evolution

(a) Derivation

Start:

$$\langle \mathbf{A} \rangle = \langle \psi | \hat{\mathbf{A}} | \psi \rangle$$

Differentiate:

$$\frac{d}{dt} \langle \mathbf{A} \rangle = \left\langle \frac{\partial \psi}{\partial t} | \mathbf{A} | \psi \right\rangle + \langle \psi | \mathbf{A} | \frac{\partial \psi}{\partial t} \rangle + \langle \psi | \frac{\partial \mathbf{A}}{\partial t} | \psi \rangle$$

Use Schrödinger equation:

$$i\hbar \frac{\partial \psi}{\partial t} = \mathbf{H} \psi$$

Substitute:

$$\frac{d\langle \mathbf{A} \rangle}{dt} = \frac{i}{\hbar} \langle [\hat{\mathbf{H}}, \hat{\mathbf{A}}] \rangle + \left\langle \frac{\partial \mathbf{A}}{\partial t} \right\rangle$$

(b) Apply to position

$$\hat{\mathbf{H}} = \frac{p^2}{2m}$$

$$\hat{\mathbf{H}}, x = \frac{1}{2m} [p^2, x]$$

Use:

$$[\hat{p}, x] = -i\hbar$$

Result:

$$\frac{d\langle x \rangle}{dt} = \frac{\langle p \rangle}{m}$$

Velocity operator emerges

(c) Conservation condition

If:

$$[\hat{\mathbf{H}}, \hat{\mathbf{A}}] = 0 \text{ and } \frac{\partial A}{\partial t} = 0$$

Then:

$$\frac{d\langle A \rangle}{dt} = 0$$

A is conserved

IV. Harmonic Oscillator Algebra

(a) Commutator

$$\hat{a} = \frac{1}{\sqrt{2\hbar m\omega}} (m\omega x + ip)$$

Compute:

$$[\hat{a}, \hat{a}^\dagger] = 1$$

(b) Hamiltonian

$$\hat{\mathbf{H}} = \hbar\omega(\hat{a}^\dagger\hat{a} + \frac{1}{2}) \quad (6.8)$$

(c) Orthonormality

$$\langle n | m \rangle = \delta_{nm}$$

Proof:

- Ladder-Operators generate states
- Inner-product preserved recursively

V. Advanced Systems

Angular-Momentum

(a) Lie Algebra

$$[J_i, J_j] = i\hbar\epsilon_{ijk}J_k$$

Derived from rotational symmetry generators

(b) Ladder-Operators

$$J_{\pm} = J_x \pm iJ_y$$

$$[J_z, J_{\pm}] = \pm\hbar J_{\pm}$$

(c) Eigenvalues

$$J^2 |jm\rangle = \hbar^2 j(j+1) |jm\rangle$$

$$J_z |jm\rangle = \hbar m |jm\rangle$$

VI. Perturbation-Theory

(a) First Order

$$E_n^{(1)} = \langle n^{(0)} | V | n^{(0)} \rangle$$

(b) Second Order

$$E_n^{(2)} = \sum_{k \neq n} \frac{|\langle k | V | n \rangle|^2}{E_n^{(0)} - E_k^{(0)}}$$

(c) Degeneracy

- Degenerate-states mix

- Solve matrix:

$$V_{ij} = \langle i | V | j \rangle$$

Diagonalize within subspace

VII. Conceptual

Decoherence

- System interacts with environment

- Phase information lost
- Superposition(classical mixture)

Explains Classical world emergence

VIII. Path-Integral

Core Idea

$$\text{Amplitude} = \sum_{\text{paths}} e^{\frac{i}{\hbar} S[x(t)]}$$

Action

$$S = \int L \, dt$$

Classical-Limit

- Does not account for a dominant path/stationary action
- Quantum Mechanics recovers Classical Mechanics

Quantum Mechanics becomes Linear Algebra, Operator Theory, and Symmetry, NOT just formulas.

Chapter 7 Example Functions

7.0.1 Equations

Plane-waves $\longrightarrow$

$\psi(x, t) = e^{(ikx - i\omega t)}$; we then differentiate this function to obtain the Energy, and Momentum operator, using the Chain Rule...and applying Plancks' rule(multiplying by $i\hbar$).

$i\hbar \frac{\partial}{\partial t} \psi(x, t) = i\omega e^{(ikx - i\omega t)} = i\omega \psi(x, t)$, seeing the eigenfunction on the left side of the equation(taking the derivative), and the resulting constant eigenvalue on the right hand side of the equation.

Energy-operator($\hat{E}$) $\rightarrow$

$$i\hbar \frac{\partial}{\partial t} \psi(x, t) = i\hbar(-i)\omega \psi(x, t) = \hbar\omega \psi(x, t) \quad (7.1)$$

Momentum operator($\hat{P}$) $\rightarrow$

$$(-i\hbar) \frac{\partial}{\partial x} \psi(x, t) = (-i\hbar)ike^{ikx - i\omega t} = \hbar k \psi(x, t) \quad (7.2)$$

(Note -i cancels i), $k = \frac{2\pi}{\lambda}$; $\omega = kc$

Vector and Matrix multiplication $\rightarrow A\vec{v} = \lambda \vec{v}$, where λ is an eigenvalue, and $\vec{v}$ is the eigenvector.

Energy of a photon $\rightarrow E = \hbar\omega$; $\vec{P} = \hbar k$; $\Delta m = 0$ (m is for mass); $E = Pc$

Discrete Fourier Transform $\rightarrow$

Consider the space of 2π periodic functions $f: \mathbb{R} \rightarrow \mathbb{C}$, with a continuous 1st derivative; We have $E_k(x) = e^{ikx}$, $k \in \mathbb{Z}$. Using Eulers' formula($e^{i\alpha} = \cos \alpha + i \sin \alpha$); $e^{2\pi i} = 1$, $E_k(x)$ by substitution, $\equiv E_k(x) = \cos(kx) + i \sin(kx)$; Here E_k is an orthonormal set.

Now, for the Hermitian Scalar Product $\langle f | \frac{1}{2\pi} \int_0^{2\pi} \overline{f(x)} \cdot g(x) dx$, we have $\langle E_k | E_s \rangle = \frac{1}{2\pi} \int_0^{2\pi} e^{-ikx} \cdot e^{isx} = \frac{1}{2\pi} \int_0^{2\pi} e^{i(s-k)x} dx$ (factored), $= \frac{1}{2\pi} \frac{1}{i(s-k)} e^{i(s-k)x} \Big|_{x=0}^{x=2\pi}$.

Note that $\iff s \text{ and } k = 0, \begin{cases} 0, \text{ if } s \neq k \\ 1, \text{ if } s = k \end{cases}$

Since its periodic($\frac{1}{2\pi} \int_0^{2\pi} 1 dx = 1$, if $s = k$); With the resulting anti-derivative of the function. So, we can pass from $(f_0, f_1, \dots, f_{n-1}) \rightarrow$ Fourier coefficients $(c_0, c_1, \dots, c_{n-1})$. This is the Discrete Fourier series; A discrete signal capturing frequency information, given that $C_k = \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} f_j e^{\frac{i2\pi kj}{N}}$ (DFT); Note that $C_k = \langle E_k | f \rangle = \sum_{j=0}^{N-1} \overline{E_{kj}} f_j$, where N is the number of samples.

Inverse Discrete Fourier Transform(IDFT) $\rightarrow$

$$f_j = \sum_{k=0}^{N-1} C_k E_{kj} = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} C_k e^{-\frac{i2\pi kj}{N}}.$$

Langranges' Theorem $\rightarrow$

Let G be a group of an order $|G|_n$; Then, $\forall a \in G, a^n = e$; So, $\mathbb{Z}_p^* = G$. It has a multiplicative inverse.

Fermats' Little Theorem $\rightarrow$ Let p be a prime; Then $\forall a \in \mathbb{Z}_p^*, a^{p-1} = 1 \pmod p$, (a) is not divisible by p .

In an RSA setting, $m = p \cdot q$, p and q are Primes; For $n = |\mathbb{Z}_m^*| = (p-1) \cdot (q-1)$; If (a) is not divisible by p or q , then $a^{(p-1)(q-1)} = 1 \pmod{pq}$.

Hadamard $\rightarrow H; |0\rangle \rightarrow$

$$\frac{1}{\sqrt{2}} |0\rangle + |1\rangle, \text{ and } |1\rangle \rightarrow -\frac{1}{\sqrt{2}} |0\rangle + |1\rangle$$

$$H = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

Eulers' Identity $\rightarrow e^{i\pi} = -1$

Adiabatic-approximation $\rightarrow \Psi(t) = e^{i\theta_k(t)} e^{i\gamma_k(t)} | \psi_k(t) \rangle$

Lowest-Landau-level $\rightarrow$

$$\Psi_m(z) \propto z^m e^{-|z|^2} \quad (7.3)$$

Remark Quantum Fourier Transform $\rightarrow U_{QFT}$ is not a Hermitian operator, $U_{QFT} \neq U_{QFT}^\dagger$. It creates an equally weighted superposition of states, leaving amplitudes as complex numbers, while Hadamard leaves them as $\mathbb{R}$. QFT also allows derivation of Fourier Basis, used in Quantum Mechanics in order to ease and describe various calculations involved in the Theory.

Baker-Campbell-Hausdorff Theorem $\rightarrow$

$$\log(e^X e^Y) = X + \int_0^1 g(e^{\text{ad } X} e^{t \text{ad } Y})(Y) dt. \quad (7.4)$$

For a complete derivation, see Dynkin's analysis of Lie groups.

Perturbed-System $\rightarrow$

$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{1}{2}m^2\omega^2 x^2 - \frac{\lambda}{2}x^2, \text{ with an exact solution of } E \approx \frac{\hbar}{2} - \frac{\hbar\lambda}{4m\omega}$$

Perturbation-Theory $\rightarrow$

$$E = E^{(0)} + \langle \psi^{(0)} | \lambda \hat{w} | \psi^{(0)} \rangle = \frac{\hbar\omega}{2} + \langle \psi^{(0)} | -(\frac{\lambda}{2})x^2 | \psi^{(0)} \rangle = \frac{\hbar\omega}{2} - \lambda \frac{\lambda\hbar}{4m\omega} = -\frac{\lambda}{2} \langle x^2 \rangle = -\frac{\lambda}{2} \sigma^2. \quad (7.5)$$

Note that $\psi^{(0)}(x) = ce^{-\frac{x^2}{4\sigma^2}}$ and $\sigma^2 = \frac{\hbar}{2m\omega}$.

Cross-Product $\rightarrow$

The cross product of $\mathbf{u} = (u_1, u_2, u_3)$ and $\mathbf{v} = (v_1, v_2, v_3)$ is a vector $(\mathbf{u} \times \mathbf{v}) =$

$$\begin{pmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{pmatrix} = (u_2 v_3 - u_3 v_2) \mathbf{i} + (u_3 v_1 - v_1 u_3) \mathbf{j} + (u_1 v_2 - u_2 v_1) \mathbf{k}.$$

The vector $(\mathbf{u} \times \mathbf{v})$ is perpendicular to $\mathbf{u}$ and $\mathbf{v}$.

The cross product ($\mathbf{v} \times \mathbf{u}$) is $-(\mathbf{u} \times \mathbf{v})$.

Note that $(\mathbf{u} \times \mathbf{u}) = 0$,

$\|\mathbf{u} \times \mathbf{v}\| = \|\mathbf{u}\| \|\mathbf{v}\| |\sin \theta|$, and $|\mathbf{u} \cdot \mathbf{v}| = \|\mathbf{u}\| \|\mathbf{v}\| |\cos \theta|$.

Time evolution Hamiltonian operator, form a complete basis $\rightarrow$

$$i \frac{d}{dt} |\psi\rangle(t) = i \frac{d}{dt} \hat{U}(t) |\psi(0)\rangle = \hat{H} \hat{U}(t) |\psi(0)\rangle = \hat{H} |\psi(t)\rangle \quad (7.6)$$

Note that $\frac{d}{dt} \hat{U}(t) = -\frac{i}{\hbar} \hat{H} \hat{U}(t)$, its like the Schrödinger equation!, and $\hat{U}(t)|\psi_n\rangle = e^{\frac{-iE_n t}{\hbar}} |\psi_n\rangle$ too.

Quantum-jump $\rightarrow$

An abrupt projection of a system to an Eigenstate ψ_0 , with a dissipative process(spontaneous emission).

The evolution of the system starts from zero, ruled by the effective Hamiltonian. The modification of $\psi(t)$ by **non-observation** of spontaneous emission, reduces the population of the state excited by $1 - \frac{1}{2}\Gamma dt$, while the ground-state remains unchanged. Every **quantum jump** projecting the system into the ground-state constitutes a **measurement**.

Monte Carlo Wave-function(MCWF) simulation $\rightarrow$

$$\psi(t) \mapsto |\psi(t+dt)\rangle \equiv \begin{cases} \frac{(1-i\hat{H}dt)|\psi(t+dt)\rangle}{\sqrt{\langle\psi(t)|\psi(t)\rangle}} & \text{if } \zeta > 1 - \langle\psi(t)|\psi(t)\rangle \\ |\psi_0\rangle & \text{if } \zeta < 1 - \langle\psi(t)|\psi(t)\rangle \end{cases}.$$

Concept of Magnitude $\rightarrow$ If $w = 3 + 4i$, then $|w| = \sqrt{\Re(w^2) + \Im(w^2)} = \sqrt{3^2 + 4^2} = 5$

Cube roots of 1 $\rightarrow$

$$z^3 = 1 \implies (x + iy)^3 = 1 \implies x^3 + 3x^2(iy) + 3x(iy)^2 + (iy)^3 = 1 \implies (x^3 - 3xy^2) + i(3x^2y - y^3) = 1;$$

$$\text{So, } x^3 - 3xy^2 = 1, \text{ and } 3x^2y - y^3 = 0.$$

$x = 1, y = 0$, and $z = 1$, is a solution.

If x and y are allowed to be complex, then there are three solutions (if x is non-zero). It has three cube roots. A real number has one real cube root, and two further cube roots which form a complex conjugate pair. For instance, the cube roots of 1 are: $1, \frac{1}{2} + \frac{\sqrt{3}}{2}i, -\frac{1}{2} + \frac{\sqrt{3}}{2}i$.

Note that,

the principal cube root of a negative number is a complex number, instance $\sqrt[3]{-8}$ is NOT -2, but rather $(1 + i\sqrt{3})$. Quartic equations can also be solved in terms of cube roots and square roots.

Seiberg-Witten functional $\rightarrow D_{A\varphi} = 0$,

$$F_A^+ = \frac{1}{4} \langle e_j \cdot e_k \cdot \varphi, \varphi \rangle e^j \wedge e^k \quad (7.7)$$

Fundamental electron velocity of the Hydrogen states $\rightarrow$

$$E_1 = \frac{-\hbar^2}{2m_e a_B^2} \quad (7.8)$$

Relativistic velocity of the electron $\rightarrow$

$$v = \sqrt{\frac{2E_1}{m_e}} = \frac{\hbar}{m_e a_B} = \frac{e^2}{4\pi\epsilon_0\hbar} = \alpha c$$

Velocity-operator $\langle \hat{v} \rangle \rightarrow$

$$\frac{i\hbar}{m} \int \psi^* \frac{\partial \psi}{\partial x} dx$$

[Velocity as motion of expectation, applies to Momentum(p), **H**(Hamiltonian total Energy of a system), and Kinetic Energy + Potential Energy (T).

Bells' Inequality $\rightarrow |P(\mathbf{a}, \mathbf{b}) - P(\mathbf{a}, \mathbf{c})| \leq 1 + P(\mathbf{b}, \mathbf{c})$

Klein-Gordon equation $\rightarrow$

$$\frac{1}{c^2} \frac{\partial^2}{\partial t^2} - \nabla^2 + \left(\frac{m_e c}{\hbar}\right)^2 \psi = 0$$

Note that $[\partial^\mu, \partial^\nu u] = [\partial^{\mu\nu}]$. The matching-index-height rule is satisfied.

7.1 Sampled wave-function distribution

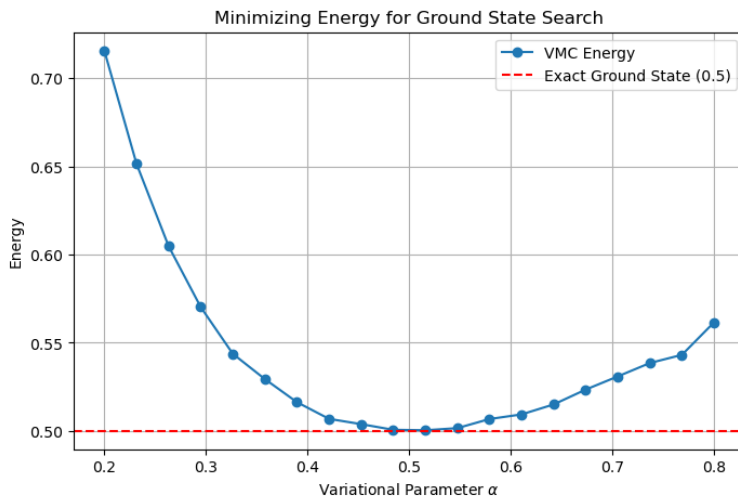


Figure 7.1: Minimizing Energy for a ground-state

[https://citizendium.org/wiki/index.php?title=Particle_in_a_box&oldid=984214]

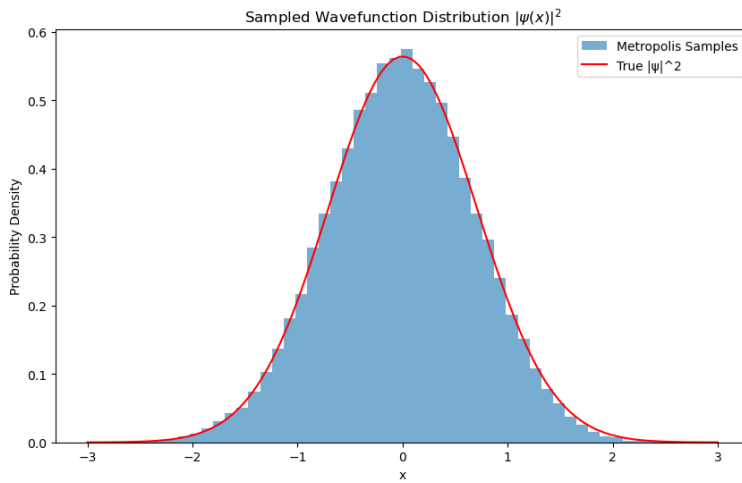


Figure 7.2: Estimated Ground State Energy, with its' analytical solution

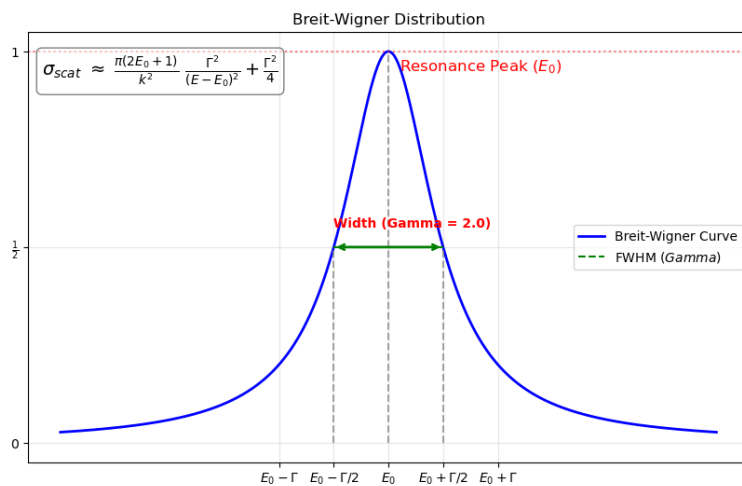


Figure 7.3: Breit-Wigner distribution

Chapter 8 Angular Momentum

The Angular Momentum of a rigid body rotating with an angular velocity ($\vec{\omega}$) about a fixed axis, and having a moment of inertia (I) matrix elements, with respect to that axis, is given by $L = I\vec{\omega}$ ($\vec{\omega}$ in rad/s). The rate of change of angular momentum of a particle equals the torque of the net force acting on it. The moment of inertia about some given axis is related to the moment about a parallel axis through the center of mass.

8.1 Rigid-Body Rotation

Any general displacement of a rigid body can be represented by a translation plus a rotation. We split the problem of rigid body motion into two separate phases, one concerned solely with the translational motion of the body, the other, with its' rotational motion. The total angular momentum divides naturally into contributions from the translation of the center of mass and, from the rotation about the center of mass.

The total angular momentum will involve the Cartesian coordinates of the center of mass, and the Euler angle coordinates. The Potential Energy in a uniform gravitational field will depend only upon the Cartesian vertical coordinate of the center of gravity. The two groups of coordinates will then be completely separated, and the translational and rotational problems can be solved independently of each other. We will link derivatives relative to a coordinate system fixed at some point in the rigid body.

The derivatives relative to any rigid body axes must vanish, and with $\vec{\omega}$ as being the angular velocity vector appropriate to the first coordinate system. The angular momentum vector is related to the angular velocity by a linear transformation. The operator (I), a Tensor of the second rank, acting upon the vector $\vec{\omega}$, results in the physically new vector (L). This rotation can be expressed in terms of the Euler angles (ϕ), (θ), and (ψ). A proper choice of these angles will transform

(I) into its diagonal form: $(I_D) = \begin{pmatrix} I_1 & 0 & 0 \\ 0 & I_2 & 0 \\ 0 & 0 & I_3 \end{pmatrix}$, where (I_1), (I_2), and (I_3), are the eigenvalues of (I), referred to as

the components of the principal moment of inertia tensor. Three roots are the desired principal moments. All directions perpendicular to the axis of symmetry are then alike, which is the mark of a double root to the equation. The principal axes are then the symmetry axis, and any two perpendicular axes in the plane normal to the symmetry axis. The principal moments of inertia cannot be negative, because as the diagonal elements in the principal axes system have the form of sums of squares. For one of the principal moments to vanish, all points of the body must be such that two coordinates of each particle are zero. See Figure 8.1

This can happen only if all points of the body are collinear with the principal axis corresponding to the zero principal moment. Any two axes perpendicular to the line of the body will then be the other principal axes. Indeed, this is a limiting case of a body with an axis of symmetry passing through the origin. If two of the roots of the equation are equal, the inertia ellipsoid thus has two equal axes, and is an ellipsoid of revolution. If all three principal moments are equal, the inertia ellipsoid is a sphere.

It is worth emphasizing that the inertia tensor (I), and all the quantities associated with it (principal axes, principal moments, inertia ellipsoid...), are only relative to some particular point fixed in the body. If the point is shifted elsewhere in the body, all the quantities will in general be changed. The $\vec{\omega}$ vector, will always move such that the corresponding normal to the inertia ellipsoid is in the direction of the angular momentum. In particular, the direction of (L) is fixed in space, and it is the inertia ellipsoid (fixed with respect to the body), that must move in space in order to preserve this connection between ω and (L). Both T , the Kinetic Energy, and L , the Angular Momentum, are constants of the motion, and the tangent plane is always a fixed distance from the origin of the ellipsoid. The total angular velocity $\vec{\omega}$ is constant in magnitude, and precesses about the (z) axis with the same frequency. The symmetry axis, is one of the principal axes, and will be chosen as the (z) axis of the coordinate system fixed in the body. Since one point is stationary, the configuration of the top is completely specified by the three Euler angles: (θ) gives the inclination of the (z) axis from the vertical, (ϕ) measures the azimuth of the top about the vertical, while (ψ) is the rotation angle of the top about its own (z) axis.

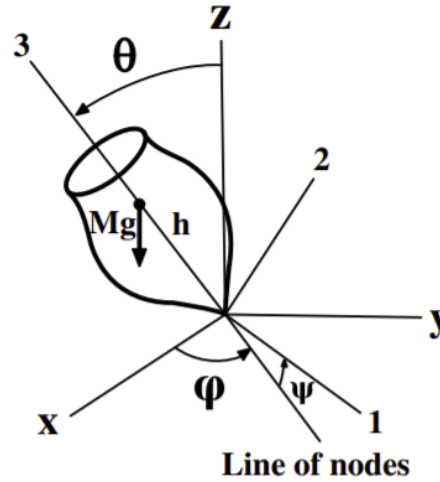


Figure 8.1: The relation between ϕ , θ , and ψ

The distance of the center of gravity (located on the symmetry axis), from the fixed point, will be denoted by (l). The angular velocity of precession, is given by $\dot{\phi} = \frac{1}{I_3 \omega_3} \frac{\partial V}{\partial (\cos \theta)}$. There are two examples of precession phenomena in nature: The types of precession that arise from the torques induced by Earth's equatorial "bulge," and the precession of moving charges in a magnetic field.

The Earth is a top whose figure axis, is precessing about the normal to the ecliptic, the plane of Earth's orbit, a motion known astronomically as the precession of the equinoxes. The precession rate, relative to the orbital angular velocity, ω_0 , is then: $\frac{\dot{\phi}}{\omega_0} = -\frac{3}{2} \frac{\omega_0}{\omega_3} \frac{I_3 - I_1}{I_3} \cos \theta$.

$\theta = 23^\circ 27'$, from the solar-induced precession, such as to cause a complete rotation of the figure axis, about the normal to the ecliptic (plane of Earth's orbit).

Magnetic moment, and the angular momentum also have a similar form. For the Classical gyromagnetic ratio, the precession angular velocity is known as the Larmor-frequency. To first order in $\mathbf{B}$, the effect of a constant magnetic field on a Classical system, is to superimpose on its normal motion, a uniform **precession**, with **angular frequency**.

Chapter 9 Spin

This is a property of the universe.

A two-particle wave-function is a difficult proposition, calculated with $\hat{H}$, integrated on all dimensions of an indistinguishable particle! See Eq. (6.8). Electrons are confined, and DON'T interact.

9.0.0.1 Quantum Projection

Now, for the contradiction of the proposal:

Suppose that we have a molecule containing two atoms in a state in which the total spin is zero, and that the spin of each atom is $\frac{\hbar}{2}$, and does not change the total angular momentum. The two atoms will begin to separate, and will soon cease to interact. Their combined spin angular momentum, however, remains equal to zero, because by hypothesis, no torques have acted on the system. If the spin were a Classical angular momentum variable, the interpretation of this process would be as follows: While the two atoms were together in the form of a molecule, each component of the angular momentum of each atom would have a definite value that was always opposite to that of the other, thus making the total angular momentum equal to zero. When the atoms separated, each atom would continue to have every component of its spin angular momentum opposite to that of the other. The two spin-angular-momentum vectors would therefore be **correlated**. These correlations were originally produced when the atoms interacted in such a way as to **form a molecule of zero total spin**, but after the atoms separate, the correlations are maintained by the deterministic equations of motion of each spin vector separately, which bring about **conservation of each component of the separate spin-angular momentum vectors**. This means that, a measurement of any component of the spin of atom No. 1 provides, as in Classical Theory, an indirect measurement of the same component of the spin of atom No. 2, without in any way disturbing that particle! For since there is no interaction with particle No. 2, the process of measurement cannot have affected this particle in any way. But now let us remember that, in each case, the observer is always free to reorient the apparatus in an arbitrary direction while the atoms are still in flight, and thus to obtain a definite (but unpredictable) value of the spin component in any direction that he chooses. Since this can be accomplished without in any way disturbing the second atom, we conclude that if ERP assumptions are applicable, precisely defined elements must exist in the second atom, corresponding to the simultaneous definition of all three components of its spin. Because the wave-function can specify, at most, only one of these components at a time with complete precision, we are then led to the conclusion that the wave-function does not provide a complete description of all elements of reality existing in the second atom. If this conclusion were valid, then we should have to look for a new Theory in terms of which a more nearly complete description was possible. We shall see, however, that the analysis given by ERP involves in an integral way the implicit assumptions that the world is actually made up of separately existing and precisely defined "elements of reality." Quantum Theory, however, implies a quite different picture of the structure of the world at the microscopic level. This picture leads, as we shall see, to a perfectly rational interpretation of the hypothetical experiment of ERP within the present framework of the Theory. Such correlations can be demonstrated, for example, in a process in which the (z) component of the spin of each atom is measured by allowing each atom to pass through a separate Stern-Gerlach apparatus. After the measurement, the system has been transformed from one that has a indefinite combined angular momentum, **The wave-function for a system of zero total spin is the same**. Both particles have a positive value together, or that both particles have a negative value together, with the constant phase factor of $[e^{\frac{\pm i\mu\mathcal{H}\alpha'\Delta t(x_1-x_2)}{\hbar}}]$, where each atom obtains an opposite momentum corresponding to its opposite spin. Because the apparatus is Classically describable, the apparatus wave-functions [which depend on the axes of (z_1) and (z_2)], multiply the spin wave-function by phase factors, so that we finally obtain: $\psi = \frac{1}{\sqrt{2}} (\psi_{x_1} e^{i\alpha x_1} + \psi_{x_2} e^{i\alpha x_2})$, where αx_1 and αx_2 are separate in the phase factor. We conclude that the value of σ_x , for each particle is also correlated to that of the other, in such a way that the sum of the two is zero. We see therefore, that the type of correlation of σ_x which can develop depends on the sign with which ψ_{x_1} and ψ_{x_2} are added, and therefore also on the combined **angular momentum**. We assert that the wave-function (in principle), can provide the most complete possible description of a system, that is consistent with the actual structure of matter. There are actually no precisely defined "elements of reality" belonging to the **electron**. The **probability** for the development of a definite value of any spin component in a suitable process of measurement

is proportional to the square of the amplitude of the coefficient of the part of the wave-function corresponding to this component. Thus, we must give up the Classical picture of a precisely defined spin variable associated with each atom, and replace it by our Quantum concept of a potentiality, the probability of whose development is given by the wave-function. As long as definite phase relations exist between $(x_1)+$ and $(x_2)-$, one cannot categorize (or classify) the system as having a spin which corresponds either entirely to $(x_1)+$ or entirely to $(x_2)-$, with respective probabilities $|x_1+|^2$, and $|x_2-|^2$. It is only when the wave-function is an eigenfunction of a given spin component, that the system is certain (in interaction with a suitable apparatus) to develop a predictable value of that spin component. The wave-function therefore describes the propagation of correlated potentialities. The concept of spin is a one-to-one correlation, and precisely define Mathematical abstractions.

9.0.0.0.2 Self-rotation of the Electron Spin is an abstract property (A small rigid particle, rotating about an axis), exceeding the speed of light (C).

Chapter 10 Spherical Harmonics

Chapter 11 Scattering

Chapter 12 Perturbation Theory

Chapter 13 Related Theorems

13.0.0.0.1 Blochs' Theorem

For periodic potentials, $V(x + a) = V(x)$; So, $\Psi(x + a) = e^{iKa}\Psi(x)$, for some constant K .

$$K = \frac{2\pi n}{a}, n = 0, \pm 1, \pm 2, \dots$$

$$K^2 = \frac{2mE}{\hbar^2}$$

$$\therefore \Psi(x) = e^{iKa} [A \sin(K(x + a)) + B(\cos(K(x + a)))]$$

13.0.0.0.2 Wicks' Theorem

13.0.0.0.3 Wigner-Eckart Theorem

13.0.0.0.4 virial Theorem See Eq. (4.4)

13.0.0.0.5 Gram-Schmidt Theorem

13.0.0.0.6 Optical Theorem

13.0.0.0.7 Parsevals' Theorem

13.0.0.0.8 Ritz Theorem

13.0.0.0.9 Steiners' Parallel-Axis Theorem

13.0.0.0.10 Larmors' Theorem

Chapter 14 Time-Dependent Problems

There's no possible state with $E = 0$, as illustrated by Figure 7.1; Thus, it is convenient to adopt periodic boundary conditions, which require that the wave-function is unaffected by a translation of any of the three Cartesian coordinates.

Chapter 15 Allowed Energies In Materials, Considering Temperature

In [http:\pages.cs.wisc.edu/kovar/hall.html](http://pages.cs.wisc.edu/kovar/hall.html),

Chapter 16 Quanglement

16.1 Entanglement

There is a need to describe systems that extend over macroscopic distances in ways that are inconsistent with Classical ideas. In Quantum Mechanics, two observable particles (x_1), and (x_2) commute by Eq. (6.3). How can measurement of particle (x_2)s' Momentum, interfere with (x_1)s', if they're far apart? The way out of this dilemma within Quantum Mechanics is to suppose that the second measurement, which gives particle 1 a definite Position, does indeed prevent particle 2 from having a definite Momentum, even though the two particles are far apart. The states of the two particles are said to be **entangled**. When two systems have wave-functions which differ by at most a constant phase factor, they are said to be in the same Quantum state. It is essential that in every measurement, there shall actually be a disturbance arising from the apparatus that destroys all elements of corresponding to observables that do not commute with the measured variable.

Chapter 17 Problems

Chapter 18 Appendix

Remark

This appendix covers some of the Mathematics used in exposition of operators, and linear equations. Here you will find the special functions that are relevant to our current application. These require algebraic techniques.

Calculus is also necessary to understand much of this material, for it is always used in advanced topics.

18.1 Rules for Continuous Functions[Apply Limit Laws]

Product Rule → Let $f(x)$ and $g(x)$ be differential functions. Then,

$$\frac{d}{dx}f(x)g(x) = \frac{d}{dx}[f(x) \cdot g(x)] + \frac{d}{dx}[g(x) \cdot f(x)] \quad (18.1)$$

That is, if $[f(x) = f(x) g(x)]$, then $[f(x) = f'(x)g(x) + g'(x)f(x)]$.

This means that the derivative of a product of two functions is the derivative of the first function times the second function, plus, the derivative of the second function times the first function.

1. Quotient Rule

2. Power Rule

Chain Rule → Let f and g be functions. $\forall x$ in the domain of g , for which g is differentiable at x , and f is differentiable at $g(x)$, the derivative of the composite function $h(x) = f(g(x))$ is given by

$$h'(x) = f'(g(x))g'(x) \quad (18.2)$$

Alternatively, if (y) is a function of (u) , and (u) is a function of (x) , then $\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx}$ e.g. $\frac{d}{dx}(2x^2 - 5)^5 = 20(x)(2x^2 - 5)^4$

Contour Integrals → If the function $f(z)$ has isolated singularities (poles) inside a closed contour (γ) , the integral around that contour is equal to $2\pi i$ times the sum of the residues at those singularities; $\oint_{\gamma} f(z)dz = 2\pi i \sum \text{Res}(f, z_k)$

18.1.1 Rank-of-a-Matrix

18.2 List of experiments

1. Franck-Hertz
2. Davisson-Germer
3. Photo-electric effect
4. Youngs double-slit
5. Stern-Gerlach
6. EPR
7. Badurek-Rauch-Tuppinger
8. SQUID(A Superconducting Quantum Interference Device (SQUID) experiment, measures extremely weak magnetic fields by exploiting Josephson junctions, and magnetic flux quantization within a superconducting loop, cooled by liquid nitrogen at 77°K or -196.1°C .
9. Delayed-choice

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